

# A Low-Fan-In Consequence Repository

Derived from *Depths of Induction and Formalized Self-Awareness*

## Strength-Ordered Consolidated Edition 0.2

182 proved catalogue entries from Volumes I–V

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## Suggested arXiv subject classification

**Primary:** cs.LO – Logic in Computer Science. **Potential cross-lists:** math.LO – Logic; cs.CC – Computational Complexity; cs.AI – Artificial Intelligence for the learned theorem-proving/search material.

*This is an authorial classification suggestion, not an arXiv assignment. The classification is included as repository metadata rather than as a claim that arXiv has assigned these categories.*

# Repository contract

This edition consolidates 182 proved catalogue entries: Volume I = 50, Volume II = 35, Volume III = 36, Volume IV = 27, and Volume V = 34. Stable theorem IDs are retained.

Each entry carries a compact provenance signature of the form: LF-XX.X | Src: \$n | Par: {...} | f:n | Hyp: ... | Stat: P | Use: ... . Here f is the displayed immediate theorem-parent count used by the source catalogue; definitions and declared elementary background are not counted as theorem vertices.

The order below is an editorial mathematical-strength ordering, not a claim about literature novelty, importance, minimum proof length, or minimum possible fan-in. It is intended to help readers find the most structurally powerful consequences first.

## Strength-ordering rubric

- **Generality:** broader quantified or system-level statements rank above fixed-instance or purely local checks.
- **Exactness:** equivalences, equality/preservation results, classifications, strict separations, and sharp transfer principles rank above one-way observations.
- **Synthesis:** results that combine or transport multiple theorem-level ingredients rank above direct restatements or definition checks.
- **Reusability:** bridge lemmas that can act as parents across sections or representations rank above isolated examples.
- **Proof substance:** nontrivial derivational arguments rank above immediate definitional consequences.

*For readability, the 182 entries are divided into four equal-size strength bands after sorting by this rubric. Within a band, stronger editorial assessments appear first.*

## Band A - strongest structural, exact, transfer, and separation results

Entries 1-46 of 182 in the strength-ordered repository. These are the entries with the strongest combination of scope, exactness, transfer value, and reusable structure.

### 1. LF-12.1. Future-complete equivalence preserves exact target distance

LF-12.1 | Src: V49 §12 | Par: {Source safe-transposition proof, including target-label preservation} | f:1 | Hyp: source hypotheses | Stat: P | Use: Strengthens a reachability-only reading of safe merging to exact unit-cost distance.

**Statement.** If  $S \equiv_{fc} S'$  under a strong bisimulation preserving target membership, then  $\text{dist}(S, W) = \text{dist}(S', W)$  for unit-cost edges, including infinite values.

**Proof.** Every length- $n$  path from  $S$  is matched step for step from  $S'$ , with equivalent endpoint and the same target-membership label. Thus an  $n$ -step target path exists from one state exactly when one exists from the other. Their sets of attainable target-path lengths, and therefore their minima, agree.

### 2. LF-C10.2. Polynomial equivalence preserves polynomially bounded proof families

LF-C10.2 | Src: Core I §10 | Par: {Source Proposition 10.10} | f:1 | Hyp: source hypotheses | Stat: P | Use: Turns proof-size upper bounds into degree-level invariants under polynomial equivalence.

**Statement.** If two proof systems are polynomially equivalent and a tautology family has polynomial-size proofs in one, then it has polynomial-size proofs in the other.

**Proof.** Use the  $p$ -simulation from the first system to the second to translate each proof with polynomial blowup. The polynomial bound composed with a polynomial translator remains polynomial.

### 3. LF-12.2. Safe quotienting preserves bounded target reachability

LF-12.2 | Src: V49 §12 | Par: {LF-12.1 and the source strong-bisimulation definition} | f:1 | Hyp: source hypotheses | Stat: P | Use: Prevents a quotient implementation from creating spurious shortcuts by switching representatives without a lifting argument.

**Statement.** For any  $n$ , equivalent states agree on whether a target is reachable within  $n$  steps. The quotient graph has the same target distance as any representative when its transitions are the induced bisimulation-class transitions.

**Proof.** The first assertion follows from equality of distances. For the second, a concrete path projects to a quotient path. A quotient path lifts from any representative: at each edge use bisimulation to match the transition that witnesses that quotient edge. Induction lifts the entire finite path, preserving length and target labels.

### 4. LF-16.1. Injective complete-state recoding preserves the transition system

LF-16.1 | Src: V49 §16 | Par: {} | f:0 | Hyp: an injective encoding  $e$  and transported transitions | Stat: P | Use: Shows precisely why exact addressing is safe as identity recoding.

**Statement.** a Let  $G$  be a labeled operational graph. Encode each complete state by  $e(s)$ , declare  $e(s) \rightarrow e(t)$  exactly a when  $s \rightarrow t$ , and transport all observation and ownership labels. Then the graph on  $e[S]$  is isomorphic to  $G$  and preserves labeled paths, unit target distances, and forced horizons.

**Proof.** Injectivity gives an inverse on the image, and the transported edge rule gives a two-way edge correspondence. Paths correspond edge by edge. Target labels and ownership are unchanged, so the same action choices and compatible plays transport controller strategies in both directions with the same lengths.

## 5. LF-20.1. Adaptive switching among sound navigation policies preserves safety

LF-20.1 | Src: V49 §20 | Par: {Source policy/legality separation} | f:1 | Hyp: source hypotheses | Stat: P | Use: Permits IAE-style adaptive control while isolating the proof-safety obligation.

**Statement.** Allow the controller to switch policies after any finite observed history. If every selected action is still checked against the same legal transition relation and every admitted certificate uses the same sound verifier, every accepted theorem remains a consequence of the current certified premises.

**Proof.** Induct on the accepted chronology. The choice of selector does not alter the legality or verification requirements of the next accepted step. The source safety argument therefore applies at every step, regardless of the switching history.

## 6. LF-33.1. Owner- and target-preserving bisimulation preserves forced horizons

LF-33.1 | Src: V49 §33 | Par: {} | f:0 | Hyp: declared definition/background | Stat: P | Use: Answers one sufficient-conditions version of the strategic-preservation question.

**Statement.** Ownership preservation is an additional explicit hypothesis. Let  $R$  be a strong bisimulation between two turn-based graphs that preserves controller/environment ownership and target membership. Matching is required in both directions for every legal successor. Then related states belong to the same finite forced-settlement stages  $A_n$ , and have equal forced horizons (including  $\infty$ ).

**Proof.** Induct on  $n$ . Target preservation gives the claim for  $A_0$ . At a controller state, a successor in  $A_n$  exists on one side exactly when a matching successor exists on the other, by bisimulation and induction. At an environment state, nonemptiness of the successor set is preserved; every successor is in  $A_n$  on one side exactly when every matching successor is on the other, using both matching directions. Membership in  $A_{n+1}$  therefore agrees. Equality of the least finite stage, or absence from all finite stages, follows.

## 7. LF-18.3. The union horizon is attained at a finite BANK stage

LF-18.3 | Src: V49 §18 | Par: {LF-18.1 and finite support as in LF-08.1} | f:2 | Hyp: source hypotheses | Stat: P | Use: Connects the limiting resource pool to an actual finite stage.

**Statement.** If  $H\text{Finf}(B_\infty, T) = h < \infty$ , some  $K$  satisfies  $H\text{Finf}(BK, T) = h$ , and every  $k \geq K$  has that same value.

**Proof.** Choose a proof attaining the finite minimum  $h$ . Its finite premise support lies in some  $BK$ , giving horizon at most  $h$  there. Since  $BK \subseteq B_\infty$ , its horizon is at least the union minimum  $h$ . The same two inequalities hold for later stages.

## 8. LF-C01.1. Inference-closed sets are exactly consequence fixed points

LF-C01.1 | Src: Core I §§1–2 | Par: {Source Theorem 1.11} | f:1 | Hyp: source hypotheses | Stat: P | Use: Turns closure checking into a fixed-point criterion for the consequence operator.

**Statement.** For every  $W \subseteq y$ ,  $W$  is inference-closed if and only if  $\text{ConF}(W) = W$ .

**Proof.** If  $W$  is inference-closed, Theorem 1.11 makes  $\text{ConF}(W)$  the least inference-closed set containing  $W$ , so it equals  $W$ . Conversely,  $\text{ConF}(W)$  is always inference-closed; if it equals  $W$ , then  $W$  is inference-closed.

## 9. LF-01.2. Strict comparison is preserved by equivalent representatives

LF-01.2 | Src: V49 §1 | Par: {LF-01.1} | f:1 | Hyp: source hypotheses | Stat: P | Use: Allows strict chains and separation results to be stored at degree level.

**Statement.** Define  $M < N$  by  $M \leq \text{clk } N$  and  $N \not\leq \text{clk } M$ . Replacing either endpoint by a mutually simulating machine preserves  $<$ .

**Proof.** Apply LF-01.1 to the forward comparison and to the reversed comparison, including its negation.

## 10. LF-C10.6. A strict family separation blocks polynomial equivalence

LF-C10.6 | Src: Core I §10 | Par: {Source Theorem 10.44 and LF-C10.5} | f:2 | Hyp: source hypotheses | Stat: P | Use: Converts a family-specific size gap into a structural non-equivalence conclusion.

**Statement.** For the Tseitin family of Theorem 10.44, the stated separation prevents resolution and Polynomial Calculus from being polynomially equivalent under the same proof-size measure.

**Proof.** LF-C10.5 gives the easy translation from resolution to Polynomial Calculus. Theorem 10.44 supplies a family for which the reverse polynomial-size simulation would contradict the stated lower/upper separation.

## 11. LF-I03.1. Rule-preserved properties hold throughout the consequence closure

LF-I03.1 | Src: V46.1 induction supplement | Par: {Source Supplement Theorem 5} | f:1 | Hyp: source hypotheses | Stat: P | Use: Provides a generic invariant engine for proof systems.

**Statement.** If all premises in  $\Gamma$  have property  $P$  and every inference rule preserves  $P$ , then every member of  $\text{ConF}(\Gamma)$  has  $P$ .

**Proof.** Supplement Theorem 5 is rule induction on finite derivations.

## 12. LF-C02.2. Canonical enumeration and breadth-first search agree on theoremhood

LF-C02.2 | Src: Core I §§1–2 | Par: {Source Theorems 2.12 and 2.17} | f:2 | Hyp: source hypotheses | Stat:  $P$  | Use: Shows two different operational chronologies can enumerate the same deductive closure.

**Statement.** For inference-closed  $Y$  and  $\Gamma \subseteq Y$ , a formula  $\phi \in Y$  eventually appears in the canonical rule enumerator  $\text{EF}, Y$  if and only if it lies in some breadth-first proof layer.

**Proof.** Theorem 2.17 says the union of enumerator states is  $\text{ConF}(\Gamma) \cap Y$ . Theorem 2.12 says the union of breadth-first layers is the same set.

## 13. LF-C04.3. Recoding preserves exact proof layers

LF-C04.3 | Src: Core I §§3–4 | Par: {LF-C04.2} | f:1 | Hyp: source hypotheses | Stat:  $P$  | Use: Allows breadth-first layers to be transported without distortion.

**Statement.** For every  $m$ ,  $t[\text{LF}(\Gamma, m)] = \text{LF } t(t[\Gamma], m)$ .

**Proof.** Membership in the left side means exact minimum proof length  $m$  before recoding. LF-C04.2 preserves that minimum exactly and  $t$  is bijective on the recoded syntax.

## 14. LF-C04.2. Recoding preserves exact shortest-proof length

LF-C04.2 | Src: Core I §§3–4 | Par: {Source Theorem 4.8} | f:1 | Hyp: source hypotheses | Stat:  $P$  | Use: Blocks a claimed proof-length speedup obtained only by invertible renaming.

**Statement.** Under the same hypotheses,  $\|t(\phi)\|_F t(t[\Gamma]) = \|\phi\|_F(\Gamma)$ .

**Proof.** A proof of either side transports to a proof of the same length on the other side. Taking minima in both directions gives equality.

## 15. LF-C10.3. Resolution lower bounds and extended-Frege upper bounds yield a two-system size separation

LF-C10.3 | Src: Core I §10 | Par: {Source Theorems 10.17 and 10.18} | f:2 | Hyp: source hypotheses | Stat:  $P$  | Use: Provides an unconditional source of proof-size speedup without claiming a wall-clock ATP theorem.

**Statement.** For the pigeonhole family covered by the source theorems, resolution requires the stated exponential-size proofs while extended Frege has polynomial-size proofs; hence the family separates the two systems by proof size.

**Proof.** Combine the lower bound of Theorem 10.17 with the upper bound of Theorem 10.18 on the same family.

## 16. LF-C01.2. Intersections of consequence fixed points are fixed points

LF-C01.2 | Src: Core I §§1–2 | Par: {Source Theorem 1.14(3) and LF-C01.1} | f:2 | Hyp: source hypotheses | Stat:  $P$  | Use: Provides a meet operation for closure invariants even when other simulation orders fail to be lattices.

**Statement.** If every  $W_i$  satisfies  $\text{ConF}(W_i) = W_i$ , then  $\text{ConF}(\bigcap W_i) = \bigcap W_i$ .

**Proof.** By LF-C01.1 every  $W_i$  is inference-closed. Theorem 1.14(3) says their intersection is inference-closed. Apply LF-C01.1 again.

## 17. LF-C04.1. Recoding preserves derivability

LF-C04.1 | Src: Core I §§3–4 | Par: {Source Theorem 4.8} | f:1 | Hyp: source hypotheses | Stat:  $P$  | Use: Establishes theoremhood invariance under pure renaming.

**Statement.** Under an invertible syntactic recoding  $t$ ,  $\Gamma \vdash_F \phi$  if and only if  $t[\Gamma] \vdash_F t(\phi)$ .

**Proof.** Theorem 4.8 transports proofs line by line in both directions using the invertibility of the recoding.

## 18. LF-C03.3. Clocked mutual simulation is transitive

LF-C03.3 | Src: Core I §§3–4 | Par: {Source Proposition 3.19} | f:1 | Hyp: source hypotheses | Stat:  $P$  | Use: Justifies treating mutual clocked simulation as a degree equivalence relation.

**Statement.** If  $M \sim N$  and  $N \sim P$ , then  $M \sim P$ .

**Proof.** Compose the forward simulations  $M \leq N \leq P$  and separately the reverse simulations  $P \leq N \leq M$  using Proposition 3.19.

## 19. LF-S21.5. Degeneracy does not descend to admissible equivalence classes

LF-S21.5 | Src: Core II §§20–21 | Par: {Source Theorem 21.7 and Corollary 21.8} | f:2 | Hyp: source hypotheses | Stat:  $P$  | Use: Warns against treating every presentation-level predicate as a degree invariant.

**Statement.** There are admissibly self-awareness-equivalent representatives that disagree on the source nondegeneracy property.

**Proof.** Theorem 21.7 constructs handed equivalent representatives; Corollary 21.8 extracts failure of invariance.

## 20. LF-I04.4. State invariants compose over any finite run prefix

LF-I04.4 | Src: V46.1 induction supplement | Par: {Source Supplement Theorem 11} | f:1 | Hyp: source hypotheses | Stat: P | Use: Supports invariants about whole search states, not only individual formulas.

**Statement.** If  $Q$  holds at the initial state and every ATP transition preserves  $Q$ , then  $Q$  holds after any finite number of stages.

**Proof.** This is ordinary stage induction supplied by Supplement Theorem 11.

## 21. LF-D18.5. Equivalence-invariant properties obey parity across equivalent representatives

LF-D18.5 | Src: Core II §18 | Par: {Source Proposition 18.21} | f:1 | Hyp: source hypotheses | Stat: P | Use: Provides a reusable test for which proposed witness properties descend to quotient classes.

**Statement.** For any property  $P$  satisfying the invariance hypothesis of Proposition 18.21, equivalent tagged systems agree on  $P$ .

**Proof.** This is the parity statement of Proposition 18.21.

## 22. LF-18.1. Conservation under a countable increasing BANK union

LF-18.1 | Src: V49 §18 | Par: {Source logical road-building theorem and LF-08.1} | f:2 | Hyp: source hypotheses | Stat: P | Use: Extends conservation from each finite deposit stage to the entire accumulated premise pool.

**Statement.** Let  $B_0 \subseteq B_1 \subseteq \dots$  with  $B_{k+1} \subseteq \text{ConF}(B_k)$ , and put  $B_\infty = S \text{ } k B_k$ . Then  $\text{ConF}(B_\infty) = \text{ConF}(B_0)$ .

**Proof.** Repeated road-building gives  $\text{ConF}(B_k) = \text{ConF}(B_0)$  for every finite  $k$ . Any proof from the union uses finitely many union premises by LF-08.1; all lie in a single  $B_k$  because the BANKs increase. Its conclusion therefore lies in  $\text{ConF}(B_0)$ . The reverse inclusion follows from  $B_0 \subseteq B_\infty$ .

## 23. LF-TM.B6. Turing equivalence is a statement about effective SICs, not all SICs

LF-TM.B6 | Src: Corrected App. B | Par: {LF-TM.B4 and LF-TM.B5} | f:2 | Hyp: source hypotheses | Stat: P | Use: States the exact boundary of the computation analogy.

**Statement.** The corrected SIC–Turing correspondence cannot be extended to all set-theoretic SICs without additional hypotheses.

**Proof.** LF-TM.B4 gives the correspondence on effective presentations; LF-TM.B5 supplies SICs outside the computable class. A universal extension would contradict that separation.

## 24. LF-03.1. Exact join criterion in the finite static subclass

LF-03.1 | Src: V49 §3 | Par: {Source characterization  $S(h, u) \leq \text{clk } S(h', u') \iff h \leq h', u \leq u'$ } | f:1 | Hyp: source hypotheses | Stat: P | Use: Turns the source's six-state obstruction into a classification for finite static joins.

**Statement.** Work within the finite static constant-halting subclass, where  $h, u \geq 0$  and  $h + u$  is a positive power of two. For a nonempty finite collection  $S(h_i, u_i)$ , put  $a = \max_i h_i$  and  $b = \max_i u_i$ . Its join in this subclass exists exactly when  $a + b$  is a power of two; in that case the join is  $S(a, b)$ .

**Proof.** If  $a + b$  is a power of two,  $S(a, b)$  is an allowed machine. The coordinate criterion makes it an upper bound below every other upper bound. Conversely, let  $P$  be the least power of two with  $P \geq a + b$ . Both  $U = S(a, P - a)$  and  $V = S(P - b, b)$  are allowed upper bounds. Any join  $J = S(h, u)$  must lie below both and above all original machines. Thus  $a \leq h \leq a$  and  $b \leq u \leq b$ , so  $J = S(a, b)$ . It is allowed only when  $a + b$  is a power of two.

## 25. LF-C05.2. Proof-covering branches recover completeness without making every branch complete

LF-C05.2 | Src: Core I §§5–6 | Par: {Source Theorem 5.7} | f:1 | Hyp: source hypotheses | Stat: P | Use: Clarifies why heuristic nondeterminism can be complete as an ensemble.

**Statement.** Under proof-covering search, every provable target occurs on at least one branch even if many individual branches miss it forever.

**Proof.** This is the content of the Branch-Covering Theorem: coverage is a property of the branch family, not of every member.

## 26. LF-I06.1. A slowly changing rank gives a lower bound on birth stage

LF-I06.1 | Src: V46.1 induction supplement | Par: {Source Supplement Theorem 15} | f:1 | Hyp: source hypotheses | Stat: P | Use: Provides a reusable way to prove search-depth lower bounds without enumerating all proofs.

**Statement.** Under the theorem's rank-increment hypothesis, a target whose rank exceeds the initial bound by  $R$  cannot be born before the rank-budget lower bound supplied by Theorem 15.

**Proof.** Along each inference layer the rank increases by at most the declared increment, so reaching rank  $R$  requires enough layers to accumulate that increase.

## 27. LF-C07.1. Conservative macro compilation preserves the theorem set

LF-C07.1 | Src: Core I §§7–9 | Par: {Source Theorem 7.12} | f:1 | Hyp: source hypotheses | Stat: P | Use: Makes a precise distinction between deductive strength and route shortening.

**Statement.** Under the hypotheses of conservative compilation, adjoining finitely many derived macro rules changes proof access but not  $\text{ConF}(\Gamma)$ .

**Proof.** Theorem 7.12 proves the compiled system conservative: each macro application expands to an original derivation, while original proofs remain legal.

## 28. LF-I05.1. Every bounded-length proof appears in a bounded closure layer

LF-I05.1 | Src: V46.1 induction supplement | Par: {Source Supplement Proposition 13} | f:1 | Hyp: source hypotheses | Stat: P | Use: Connects syntactic certificate length with a closure-stage upper bound.

**Statement.** If  $\varphi$  has a proof from  $\Gamma$  of length at most  $m$ , then it is present by the finite proof layer guaranteed by Proposition 13.

**Proof.** Proposition 13 relates proof length to the finite iterative closure construction.

## 29. LF-ML2.1. Right censoring yields a lower bound, not an exact unguided cost

LF-ML2.1 | Src: V48 Int. A | Par: {} | f:0 | Hyp: frozen empirical record | Stat: P | Use: Preserves the experiment's finite, right-censored interpretation.

**Statement.** From  $\text{EML}(\text{prcom}) = 2633$  and  $\text{Eunguided}(\text{prcom}) > 5000$ , one may conclude only  $\text{Eunguided} / \text{EML} > 5000/2633 \approx 1.90$ , not the exact ratio.

**Proof.** The unguided run stopped at the budget while still unknown, so its first-proof expansion count is known only to exceed the cutoff.

## 30. LF-02.2. Unbounded finite width and failure of well-quasi-ordering

LF-02.2 | Src: V49 §2 | Par: {Source infinite-antichain theorem} | f:1 | Hyp: source hypotheses | Stat: P | Use: Identifies the specific order-theoretic obstruction supplied by the antichain.

**Statement.** For every  $m \geq 1$  there are  $m$  pairwise incomparable finitely describable degrees. The degree order on finitely describable machines is not a well-quasi-order.

**Proof.** Take  $m$  distinct elements of the source infinite antichain. An enumeration of that antichain is an infinite sequence with no pair  $i < j$  satisfying  $a_i \leq a_j$ , violating the defining sequence condition for a well-quasi-order.

## 31. LF-SA16.2. Admissible self-awareness comparison is a preorder

LF-SA16.2 | Src: Core II §§16–17 | Par: {Source Propositions 16.9 and 16.10} | f:2 | Hyp: source hypotheses | Stat: P | Use: Provides the order-theoretic foundation for quotient degrees.

**Statement.** The admissible comparison is reflexive and transitive.

**Proof.** Proposition 16.9 supplies identity and closure under composition of admissible translations; Proposition 16.10 lifts those facts to reflexivity and transitivity of the comparison relation.

## 32. LF-31.1. Permuting a finite breadth-first layer preserves first successful depth

LF-31.1 | Src: V49 §31 | Par: {} | f:0 | Hyp: complete layer processing assumed | Stat: P | Use: Separates represented proof depth from within-layer discovery offset.

**Statement.** Suppose two searches expose exactly the same finite candidate histories in each depth layer, finish every shallower layer before the next, and differ only by permutations within each layer. If the least successful depth is  $d$ , both searches first succeed at depth  $d$ .

**Proof.** No candidate in a depth below  $d$  succeeds, so either search must reach layer  $d$ . Each completely processes that finite layer unless it succeeds earlier within the layer; because it contains a successful candidate, both succeed there.

## 33. LF-C09.2. Undecidable consequence problems exclude computable finite-effective SICs

LF-C09.2 | Src: Core I §§7–9 | Par: {Source Corollary 9.14} | f:1 | Hyp: source hypotheses | Stat: P | Use: Marks the computability boundary of finite closure search.

**Statement.** If the consequence problem on an effectively presented domain is undecidable, then no computable deductively finitely effective SIC can solve it in the sense of the source framework.

**Proof.** This is the contrapositive form recorded in Corollary 9.14: a computable finite-effective SIC would decide consequence by waiting for its finite completion stage.

## 34. LF-10.1. The append-only derivation-state graph is acyclic

LF-10.1 | Src: V49 §10 | Par: {Source operational-lifting theorem} | f:1 | Hyp: source hypotheses | Stat: P | Use: Supplies a genuine DAG for One-Step DAG mining.

**Statement.** For a fixed finite initial sequence of length  $m$ , every path of  $n$  transitions ends at a sequence of length  $m + n$ . Consequently the lifted graph has no directed cycles, and any reachable exact sequence state has a uniquely determined path length from the root.

**Proof.** Each lifted transition appends one formula. Induction gives the length formula. A nonempty directed cycle would strictly increase sequence length and return to the same sequence, which is impossible.

### 35. LF-I05.2. The omega union of finite closure stages equals the consequence closure

LF-I05.2 | Src: V46.1 induction supplement | Par: {Source Supplement Corollary 14} | f:1 | Hyp: source hypotheses | Stat: P | Use: Explains why no separate transfinite stage is needed for ordinary finitary proof closure.

**Statement.** Under the stated inference-closed-domain hypotheses, the union of all finite exhaustive stages is exactly  $\text{ConF}(\Gamma)$ .

**Proof.** Every stage is sound, while Proposition 13 ensures each finite proof appears at some finite stage; Corollary 14 records equality.

### 36. LF-A2.4. Conservative cooperation preserves dependence and independence status

LF-A2.4 | Src: V48 Int. B §§1–3 | Par: {Source Corollary 2.3} | f:1 | Hyp: source hypotheses | Stat: P | Use: Shows that conservative BANK growth cannot settle an actually different object-theory question by changing premises.

**Statement.** Under the same hypotheses,  $\text{DepF}, \text{Bk}(\Gamma)$  and  $\text{IndF}, \text{Bk}(\Gamma)$  are stage invariant.

**Proof.** Both are defined from the two derivability predicates whose truth values are constant by Corollary 2.3.

### 37. LF-C02.3. The corrected target-search machine freezes without changing first hit

LF-C02.3 | Src: Core I §§1–2 | Par: {Corrected Proof-Horizon conclusion C37} | f:1 | Hyp: source hypotheses | Stat: P | Use: Isolates a SIC-axiom repair from the proof-horizon quantity itself.

**Statement.** For a provable target  $\varphi$ , freezing the target-search breadth-first SIC at its first hit preserves the first halting stage  $\|\varphi\|_F(\Gamma)$ .

**Proof.** C37 changes only post-halt state evolution: before the target is reached, the corrected state agrees with ordinary breadth-first search, and at the first hit it freezes.

### 38. LF-A2.1. Any accepted transition invariant holds at every global stage

LF-A2.1 | Src: V48 Int. B §§1–3 | Par: {Source Theorem 2.1} | f:1 | Hyp: source hypotheses | Stat: P | Use: Provides the global analogue of single-ATP stage induction.

**Statement.** If a predicate  $Q$  holds at  $X_0$  and every accepted AMLD transition preserves  $Q$ , then  $Q(X_k)$  holds for all  $k \in \mathbb{N}$ .

**Proof.** This is Theorem 2.1 by ordinary induction on the global accepted-stage index.

### 39. LF-C06.1. Proof paths give an upper bound on theorem-graph distance

LF-C06.1 | Src: Core I §§5–6 | Par: {Source Proposition 6.8} | f:1 | Hyp: source hypotheses | Stat: P | Use: Connects certificates to graph-geometric upper bounds.

**Statement.** If a proof of  $\varphi$  from  $\Gamma$  induces a theorem-graph path of length  $n$ , then the directed graph distance from the start state to a  $\varphi$ -state is at most  $n$ .

**Proof.** Proposition 6.8 turns the proof into a legal path. A shortest path cannot be longer than a displayed path.

### 40. LF-A6.3. Resource replication restores wall-clock no-regret for a successful baseline

LF-A6.3 | Src: V48 Int. B §§6–7 | Par: {Source Proposition 6.5} | f:1 | Hyp: source hypotheses | Stat: P | Use: States exactly what extra resource assumption turns logical emulation into wall-clock dominance.

**Statement.** If  $P$  receives the same processor, memory, verifier availability, and local service rate as the successful stand-alone prover, and foreign deposits may be ignored, then AMLD first-settlement wall time is at most the stand-alone proof time.

**Proof.** Proposition 6.5 lets  $P$  reproduce the stand-alone run without resource loss; another role can only finish earlier.

### 41. LF-I08.2. Exact reflection layers make the reflection index recoverable from birth stage

LF-I08.2 | Src: V46.1 induction supplement | Par: {Source Supplement Theorem 20} | f:1 | Hyp: source hypotheses | Stat: P | Use: Provides a calibrated benchmark in which semantic reflection depth and operational stage are linked exactly.

**Statement.** For the reflection-closure machine of Theorem 20, the designated reflection formula  $\theta_k$  is born at the exact layer stated by the theorem; hence different finite  $k$  have different recorded birth layers.

**Proof.** Theorem 20 gives an exact formula for the layer as a strictly increasing function of  $k$ .

### 42. LF-ML1.2. Policy replacement can change discovery cost while leaving theoremhood invariant

LF-ML1.2 | Src: V48 Int. A | Par: {Source Proposition 1.2} | f:1 | Hyp: source hypotheses | Stat: P | Use: Separates search effort from deductive strength.

**Statement.** Two ranking policies may inspect legal candidates in different orders and therefore use different numbers of expansions before finding a proof, while accepting exactly the same class of verifier-valid certificates.

**Proof.** The proposition fixes theoremhood at the verifier boundary but leaves search order free. A permutation of a finite candidate set can move the first successful candidate to a different position.

### 43. LF-25.1. Local invariants extend to every finite accepted history

LF-25.1 | Src: V49 §25 | Par: {} | f:0 | Hyp: stated conditions + declared background | Stat: P | Use: Provides the reusable proof form for typed BANK and verifier invariants.

**Statement.** If a predicate  $I$  holds at every allowed initial state and every accepted transition preserves  $I$ , then  $I$  holds at every finitely reachable accepted state, independently of the order in which enabled transitions are selected.

**Proof.** Induct on history length. The zero-step case is the initial-state condition. For the inductive step, apply preservation to the final transition after the induction hypothesis at its predecessor.

#### 44. LF-C10.7. Correct arithmetization transfers satisfying assignments to algebraic zeros

LF-C10.7 | Src: Core I §10 | Par: {Source Proposition 10.35} | f:1 | Hyp: source hypotheses | Stat: P | Use: Provides the verifier-safe bridge between propositional satisfiability and algebraic proof systems.

**Statement.** Under the source arithmetization, a Boolean assignment satisfies the encoded propositional instance exactly when its 0–1 vector satisfies the associated polynomial constraints.

**Proof.** This is the correctness equivalence of Proposition 10.35, read in both directions.

#### 45. LF-18.2. Each finite target horizon eventually stabilizes

LF-18.2 | Src: V49 §18 | Par: {Source logical road-building theorem} | f:1 | Hyp: source hypotheses | Stat: P | Use: Gives a targetwise convergence fact.

**Statement.** Under the BANK hypotheses above, if  $\text{HFinf}(B_0, T) = h < \infty$ , then  $h_k = \text{HFinf}(B_k, T)$  eventually becomes constant and has at most  $h$  strict decreases.

**Proof.** The source theorem makes  $(h_k)$  a nonincreasing sequence of nonnegative integers starting at  $h$ . Every strict decrease lowers it by at least one. There can be at most  $h$  such decreases, after the last of which the sequence is constant.

#### 46. LF-SA16.8. The admissible quotient contains an infinite strict chain

LF-SA16.8 | Src: Core II §§16–17 | Par: {Source Corollary 16.20} | f:1 | Hyp: source hypotheses | Stat: P | Use: Shows the quotient order has genuinely nontrivial infinite height.

**Statement.** There are quotient classes  $[M_0] < [M_1] < \dots$  under the source admissible self-awareness order.

**Proof.** Corollary 16.20 supplies such a sequence using increasing reflection levels.

## Band B - strong reusable preservation, bound, closure, and simulation consequences

Entries 47–91 of 182 in the strength-ordered repository. These entries remain mathematically substantial but are generally narrower or less exact than Band A.

#### 47. LF-SA16.3. Mutual admissible comparison is an equivalence relation

LF-SA16.3 | Src: Core II §§16–17 | Par: {Source Proposition 16.12} | f:1 | Hyp: source hypotheses | Stat: P | Use: Licenses quotienting by presentation-level equivalence.

**Statement.** The relation defined by comparison in both admissible directions is reflexive, symmetric, and transitive.

**Proof.** This is Proposition 16.12 applied to the underlying preorder.

#### 48. LF-TM.B5. Not every set-theoretic SIC is computationally realizable

LF-TM.B5 | Src: Corrected App. B | Par: {Source Corollary B.20} | f:1 | Hyp: source hypotheses | Stat: P | Use: Preserves the distinction between the abstract SIC universe and executable machines.

**Statement.** Over an effectively coded domain there exist SICs that are not computable in the source sense.

**Proof.** Corollary B.20 supplies the separation from the computable/Turing-realizable subclass.

#### 49. LF-TM.B1. Configuration provability is reachability

LF-TM.B1 | Src: Corrected App. B | Par: {Source Proposition B.5} | f:1 | Hyp: source hypotheses | Stat: P | Use: Supplies a concrete bridge between proof systems and computation graphs.

**Statement.** In the configuration formal system,  $\Gamma \vdash_{\text{FT}} c$  exactly when the source reachability condition for configuration  $c$  holds from  $\Gamma$ .

**Proof.** Proposition B.5 proves the two directions between formal derivation and machine-transition reachability.

#### 50. LF-11.2. Geodesic steps decrease remaining horizon exactly by one

LF-11.2 | Src: V49 §11 | Par: {LF-11.1} | f:1 | Hyp: source hypotheses | Stat: P | Use: Gives an exact certificate target for a proposed geodesic next move.

**Statement.** If  $0 < h(S) < \infty$ , at least one successor  $U$  has  $h(U) = h(S) - 1$ . Every successor satisfies  $h(U) \geq h(S) - 1$ .

**Proof.** The finite minimum in LF-11.1 is attained because attainable finite distances are natural numbers. Its value is  $h(S) - 1$ , and all other values are at least that minimum.

## 51. LF-C03.4. A cardinality obstruction blocks synchronous equivalence in either direction

LF-C03.4 | Src: Core I §§3–4 | Par: {Source Proposition 3.11} | f:1 | Hyp: source hypotheses | Stat: P | Use: Turns state-space cardinality into a quick negative certificate.

**Statement.** If neither  $P(YM)$  injects into  $P(YN)$  nor  $P(YN)$  injects into  $P(YM)$ , then  $M$  and  $N$  cannot be synchronously equivalent.

**Proof.** Synchronous equivalence would contain a synchronous simulation in both directions. Proposition 3.11 would then supply both injections, contrary to hypothesis.

## 52. LF-C09.1. Finite inference-closed domains make exhaustive rule enumeration finitely complete

LF-C09.1 | Src: Core I §§7–9 | Par: {Source Theorem 9.7} | f:1 | Hyp: source hypotheses | Stat: P | Use: Identifies a regime where exhaustive closure is guaranteed to stabilize.

**Statement.** If  $Y$  is finite and inference-closed, the canonical rule enumerator reaches its full consequence set after finitely many stages on every input.

**Proof.** This is deductive finite effectiveness from Theorem 9.7. Monotone growth inside a finite powerset cannot add new formulas forever.

## 53. LF-C06.2. A shortest-path ATP cannot beat intrinsic graph distance

LF-C06.2 | Src: Core I §§5–6 | Par: {Source Theorem 6.9} | f:1 | Hyp: source hypotheses | Stat: P | Use: Provides a clean lower bound that is independent of heuristic ordering.

**Statement.** Under the finite/local hypotheses of the shortest-path ATP principle, no graph-search execution can first reach the target in fewer edge steps than the shortest-path distance.

**Proof.** Theorem 6.9 identifies the optimal edge-step target cost with shortest-path search. Any successful execution is itself a path and hence has length at least the minimum.

## 54. LF-A10.1. Conservative BANK growth can improve access without increasing deductive strength

LF-A10.1 | Src: V48 Int. B §§8–11 | Par: {Source Theorem 2.2 and Theorem 5.1} | f:2 | Hyp: source hypotheses | Stat: P | Use: Summarizes the central ‘road-building without new axioms’ phenomenon of the corrected AMLD integration.

**Statement.** Along conservative BANK growth, the consequence closure remains  $\text{ConF}(B_0)$  while the explicit-premise horizons for  $T$ ,  $\neg$ , and  $\perp$  are nonincreasing.

**Proof.** Theorem 2.2 gives constant consequence closure; Theorem 5.1 gives nonincreasing shortest explicit-premise horizons under BANK inclusion.

## 55. LF-SA16.9. Tag-preserving simulation cannot decrease formal level

LF-SA16.9 | Src: Core II §§16–17 | Par: {Source Proposition 16.22 and Corollary 16.23} | f:2 | Hyp: source hypotheses | Stat: P | Use: Connects operational simulation to the syntactic level hierarchy under an explicit tag bridge.

**Statement.** Under the tag-preserving simulation hypotheses of Proposition 16.22,  $\text{lev}(M) \leq \text{lev}(N)$ .

**Proof.** Proposition 16.22 preserves self-awareness statements; Corollary 16.23 records the resulting level inequality.

## 56. LF-19.1. Robust budget regions expand under safe enrichment

LF-19.1 | Src: V49 §19 | Par: {Source conditional robust-horizon monotonicity theorem} | f:1 | Hyp: source hypotheses | Stat: P | Use: Turns a pointwise monotonicity theorem into a budget-region inclusion.

**Statement.** Under that theorem’s preserved-state, controller-option, environment-response, target, and cost hypotheses, every old state satisfying  $H_{\text{force}} \leq n$  also satisfies the post-enlargement bound  $H_{\text{force}} \leq n$ .

**Proof.** The source theorem gives  $H_{\text{force}}(s) \leq H_{\text{force}}(s)$  at every formerly forced-settling state. Apply this pointwise to the region bounded by  $n$ .

## 57. LF-C10.4. Feasible interpolation turns suitable communication hardness into proof lower bounds

LF-C10.4 | Src: Core I §10 | Par: {Source Theorem 10.28} | f:1 | Hyp: source hypotheses | Stat: P | Use: Shows how a semantic/combinatorial obstruction can become a proof-complexity lower bound.

**Statement.** Whenever the hypotheses of the source interpolation lower-bound theorem are met, hardness of the associated interpolant problem rules out small proofs in the proof system under consideration.

**Proof.** This is the implication encoded by Theorem 10.28: a small proof plus feasible interpolation would yield an efficiently computable interpolant, contradicting the specified hardness premise.

## 58. LF-C03.1. Synchronous equivalence implies clocked equivalence

LF-C03.1 | Src: Core I §§3–4 | Par: {Source Proposition 3.17} | f:1 | Hyp: source hypotheses | Stat: P | Use: Records the strict hierarchy between two simulation notions.

**Statement.** If  $M \sim_{\text{syn}} N$ , then  $M \sim N$  under clocked deterministic simulation.

**Proof.** Each synchronous simulation direction is, by Proposition 3.17, a clocked simulation direction. Mutuality is therefore preserved.

## 59. LF-32.1. Order reflection suffices to transfer an antichain

LF-32.1 | Src: V49 §32 | Par: {Source infinite-antichain theorem} | f:1 | Hyp: the stated additional map hypothesis | Stat: P | Use: States a precise bridge obligation for transferring the old antichain.

**Statement.** Let  $(M_i)_{i \in \mathbb{N}}$  be the source SIC antichain, and let  $J$  assign each  $M_i$  a GBG presentation. Suppose  $J(M_i) \leq^* J(M_j) \implies M_i \leq_{\text{clk}} M_j$ . Then the presentations  $J(M_i)$  are pairwise incomparable under  $\leq^*$ .

**Proof.** Any comparison between two distinct images would reflect to a forbidden comparison in the SIC antichain. The same argument excludes the reversed comparison.

## 60. LF-A8.2. High-rank deposits require enough cumulative certificate depth

LF-A8.2 | Src: V48 Int. B §§8–11 | Par: {Source Theorem 8.1} | f:1 | Hyp: source hypotheses | Stat: P | Use: Supplies a verifier-checkable lower bound on cooperative dependency work.

**Statement.** If a formula first deposited by update  $n$  has  $r(\lambda) > r_0$ , then  $Q_n \geq \lceil (r(\lambda) - r_0)/d \rceil$ .

**Proof.** Rearrange the upper bound  $r(\lambda) \leq r_0 + dQ_n$  from Theorem 8.1 and use integrality.

## 61. LF-SA16.4. Admissible translations preserve each finite reflection stage

LF-SA16.4 | Src: Core II §§16–17 | Par: {Source Lemma 16.17} | f:1 | Hyp: source hypotheses | Stat: P | Use: Provides the bridge needed to compare reflection levels across different syntactic presentations.

**Statement.** Under an admissible translation, the image of the source reflection sequence is the corresponding reflection sequence in the target in the sense stated by Lemma 16.17.

**Proof.** The lemma proves preservation inductively across the reflection construction.

## 62. LF-A3.2. Object-level invariants are channel independent once the certificate is verified

LF-A3.2 | Src: V48 Int. B §§1–3 | Par: {Source Corollary 3.3} | f:1 | Hyp: source hypotheses | Stat: P | Use: Allows useful lemmas from a losing search role to be safely reused elsewhere.

**Statement.** Whether a valid deposited lemma preserves an object invariant depends on its accepted certificate, not on whether  $P$ ,  $R$ ,  $D$ ,  $I$ , or  $C$  happened to discover it.

**Proof.** Corollary 3.3 is exactly this channel-independence consequence of certificate-preserved birth-time induction.

## 63. LF-P20.3. Picard’s self-theory contains every finite reflection formula

LF-P20.3 | Src: Core II §§20–21 | Par: {Source Theorem 20.4 and Corollary 15.13} | f:2 | Hyp: source hypotheses | Stat: P | Use: Makes the content behind the level value explicit.

**Statement.** Under the Picard idealization, every finite reflection formula  $\theta_k$  belongs to  $\text{Self}(\text{Picard})$ .

**Proof.** Theorem 20.4 gives level  $\omega$ , hence provability of every finite  $\theta_k$  under the source definition. Corollary 15.13 places each proved reflection formula in the self-theory.

## 64. LF-01.1. Changing representatives does not change comparison

LF-01.1 | Src: V49 §1 | Par: {Source section 1 preorder property} | f:1 | Hyp: source hypotheses | Stat: P | Use: Makes a catalogue of degree comparisons independent of the chosen machine representative.

**Statement.** If  $M \sim_{\text{clk}} M'$  and  $N \sim_{\text{clk}} N'$ , then  $M \leq_{\text{clk}} N$  if and only if  $M' \leq_{\text{clk}} N'$ .

**Proof.** For the forward implication compose  $M' \leq_{\text{clk}} M \leq_{\text{clk}} N \leq_{\text{clk}} N'$ . The reverse implication uses  $M \leq_{\text{clk}} M' \leq_{\text{clk}} N' \leq_{\text{clk}} N$ . Each chain uses transitivity of the source preorder.

## 65. LF-30.1. Two-stage deterministic hitting-time bound

LF-30.1 | Src: V49 §30 | Par: {Source section 30 target-preserving hitting-time bound and LF-28.1} | f:2 | Hyp: source hypotheses | Stat: P | Use: Connects a chain of simulation certificates to a concrete operational deadline, without claiming a shortest-path bound.

**Statement.** Under the two clocked embeddings of LF-28.1, assume target preservation at both embeddings. If the source hitting stage is  $n < \infty$ , the final target is hit by stage  $\tau_2(\tau_1(n))$ .

**Proof.** Compose the state maps and target inclusions. The composite clock is bounded by LF-28.1. Apply the source hitting-time theorem to that composite witness.

## 66. LF-SA15.7. Every advertised finite level and omega is realized

LF-SA15.7 | Src: Core II §§13–15 | Par: {Source Theorem 15.32 and Corollary 15.33} | f:2 | Hyp: source hypotheses | Stat: P | Use: Shows the level scale is populated rather than merely formal notation.

**Statement.** For each  $n \in \mathbb{N}$  and for  $\omega$  as listed by the source, there exists a tagged ATP whose level is exactly that value.

**Proof.** Theorem 15.32 supplies strict finite witnesses, and Corollary 15.33 records surjectivity of the level function onto the stated range.

## 67. LF-C01.4. Exact proof layers are disjoint

LF-C01.4 | Src: Core I §§1–2 | Par: {} | f:0 | Hyp: declared definition/background | Stat: P | Use: Makes the breadth-first proof horizon into a genuine layer decomposition.

**Statement.** If  $m \neq n$ , then  $LF(\Gamma, m) \cap LF(\Gamma, n) = \emptyset$ .

**Proof.** A formula has one well-defined minimum proof length. It cannot simultaneously have two distinct minima.

## 68. LF-C10.1. p-simulation chains compose polynomial overheads

LF-C10.1 | Src: Core I §10 | Par: {Source Proposition 10.11} | f:1 | Hyp: source hypotheses | Stat: P | Use: Supports modular proof-system translations without leaving polynomial overhead.

**Statement.** A finite chain of p-simulations is again a p-simulation; the certificate-size/time overhead is bounded by a polynomial obtained by composing the finitely many polynomial translators.

**Proof.** Proposition 10.11 gives transitivity. A finite composition of polynomials is polynomial.

## 69. LF-28.1. Quantitative clock bounds compose

LF-28.1 | Src: V49 §28 | Par: {Source clocked-simulation composition} | f:1 | Hyp: source hypotheses | Stat: P | Use: Permits chains of quantitative simulations while charging their accumulated distortion.

**Statement.** Suppose  $G \leq_{\text{clk}} H$  has clock  $\rho_1(n) \leq \tau_1(n)$  and  $H \leq_{\text{clk}} K$  has clock  $\rho_2(n) \leq \tau_2(n)$ , where the bounding functions are nondecreasing on positive integers. The composite witness has clock  $\rho_2(\rho_1(n)) \leq \tau_2(\tau_1(n))$ .

**Proof.** The source composition uses  $\rho_2 \circ \rho_1$ . Bound its outer function by  $\tau_2$ , then use monotonicity of  $\tau_2$  and the bound on  $\rho_1$ .

## 70. LF-C07.2. Finite compiled workloads can be accelerated without adding theorems

LF-C07.2 | Src: Core I §§7–9 | Par: {Source Corollary 7.13 and Theorem 7.12} | f:2 | Hyp: source hypotheses | Stat: P | Use: Provides a model of safe proof engineering for a frozen finite benchmark.

**Statement.** For any finite family of already provable designated targets covered by Corollary 7.13, one may add conservative derived rules tailored to that workload while preserving the original consequence set.

**Proof.** Corollary 7.13 supplies the finite-workload compilation, and Theorem 7.12 supplies conservativity.

## 71. LF-I03.2. Two rule invariants can be conjoined

LF-I03.2 | Src: V46.1 induction supplement | Par: {Source Supplement Theorem 5} | f:1 | Hyp: source hypotheses | Stat: P | Use: Allows independent safety properties to be accumulated without increasing theorem fan-in.

**Statement.** If properties P and Q each hold on the hypotheses and are each preserved by every rule, then every consequence satisfies  $P \wedge Q$ .

**Proof.** The conjunction holds initially. A rule whose conclusion preserves both P and Q preserves their conjunction. Apply rule induction once to  $P \wedge Q$ .

## 72. LF-C03.2. Clocked simulations compose with composed clocks

LF-C03.2 | Src: Core I §§3–4 | Par: {Source Proposition 3.19} | f:1 | Hyp: source hypotheses | Stat: P | Use: Provides the basic arithmetic for multi-stage scheduler embeddings.

**Statement.** If  $(f, \rho)$  simulates M in N and  $(g, \sigma)$  simulates N in P, then  $(g \circ f, \sigma \circ \rho)$  simulates M in P.

**Proof.** This is exactly the composition construction of Proposition 3.19; injectivity and the state/halting identities compose, and the composite clock remains admissible.

## 73. LF-D18.3. A complex target-search ATP may still have level zero

LF-D18.3 | Src: Core II §18 | Par: {Corrected Proposition 18.17 and Proposition 18.18} | f:2 | Hyp: source hypotheses | Stat: P | Use: Demonstrates that proof-search sophistication and formal self-reflection are distinct axes.

**Statement.** The corrected algebraic target-search machine G can satisfy the source complexity criterion while remaining at formal self-awareness level 0.

**Proof.** Corrected Proposition 18.17 establishes the target-search/proof-system properties; Proposition 18.18 records the complexity clause together with level zero.

## 74. LF-14.1. Reachability and forced distance can have arbitrarily large gaps

LF-14.1 | Src: V49 §14 | Par: {Source min/max forced-horizon semantics} | f:1 | Hyp: source hypotheses | Stat: P | Use: Shows that the numerical gap is not bounded merely by the existence of a one-step successful route.

**Statement.** For every  $L \geq 1$  there is a finite strategic graph with  $\text{dist}(s, W) = 1$  and  $\text{Hforce}(s, W) = L$ .

**Proof.** For  $L = 1$ , use a single edge to a target. For  $L \geq 2$ , make s environment-owned with two choices: an edge directly to a target, and an edge starting a deterministic route whose total length from s to a target is L. Every play settles; the environment chooses the longer route in the worst case.

## 75. LF-S21.7. The specified transfinite calculus realizes every level up to its stated bound

LF-S21.7 | Src: Core II §§20–21 | Par: {Source Theorem 21.15} | f:1 | Hyp: source hypotheses | Stat: P | Use: Shows how the finite hierarchy can be extended only after the missing limit-stage structure is explicitly supplied.

**Statement.** For every ordinal  $\delta \leq \kappa$  covered by the theorem, there exists a tagged ATP  $M_\delta$  with the corresponding transfinite reflection behavior stated in Theorem 21.15.

**Proof.** The theorem provides the witnesses directly, conditional on the specified transfinite calculus and limit clauses.

## 76. LF-10.2. Exact proof budgets are exact reachability budgets

LF-10.2 | Src: V49 §10 | Par: {Source operational-lifting theorem} | f:1 | Hyp: source hypotheses | Stat: P | Use: Allows a target-specific mining request to specify an auditable inference budget.

**Statement.** For  $n \geq 0$ ,  $T$  has a proof using at most  $n$  primitive inference applications from  $B_0$  exactly when the lifted target region is reachable in at most  $n$  transitions.

**Proof.** Lift a proof line by line, or read the inference labels on a path. The correspondence preserves the number of primitive inference applications.

## 77. LF-C03.5. Effective non-simulation follows from ordinary non-simulation

LF-C03.5 | Src: Core I §§3–4 | Par: {Source Proposition 3.26} | f:1 | Hyp: source hypotheses | Stat: P | Use: Lets unrestricted obstructions be reused in the effective setting.

**Statement.** If  $M \leq N$ , then no effective clocked simulation  $M \leq_{\text{eff}} N$  exists.

**Proof.** Proposition 3.26 states that effective clocked simulation implies ordinary clocked simulation. Contraposition gives the claim.

## 78. LF-A8.3. Causal depth gives a sharper target-specific rank bound

LF-A8.3 | Src: V48 Int. B §§8–11 | Par: {Source Theorem 8.3} | f:1 | Hyp: source hypotheses | Stat: P | Use: Provides a schedule-insensitive lower bound focused on the target’s causal certificate structure.

**Statement.** Every deposited formula satisfies  $r(\lambda) \leq r_0 + d\delta(\lambda)$ , hence  $\delta(\lambda) \geq \lceil \max\{0, r(\lambda) - r_0\} / d \rceil$ .

**Proof.** Theorem 8.3 follows only the actual recursively expanded dependency ancestry of  $\lambda$  rather than charging every earlier deposit.

## 79. LF-30.2. Affine distortion remains affine under composition

LF-30.2 | Src: V49 §30 | Par: {LF-30.1} | f:1 | Hyp: source hypotheses | Stat: P | Use: Exposes the constants that a qualitative degree comparison leaves unspecified.

**Statement.** If  $\tau_1(n) = an + b$  and  $\tau_2(n) = cn + d$ , with nonnegative coefficients and valid nondecreasing integer-valued bounds, the final hitting stage is at most  $ca n + cb + d$ .

**Proof.** Substitute the affine expressions into  $\tau_2(\tau_1(n))$ .

## 80. LF-17.1. Bounded-latency scheduling gives a quantitative emulation bound

LF-17.1 | Src: V49 §17 | Par: {Source scheduler-robust emulation argument} | f:1 | Hyp: the modified service condition stated here | Stat: P | Use: Generalizes the source unit-rate guarantee to a practical service-latency contract.

**Statement.** Suppose a baseline-preserving prover needs  $m \geq 1$  local transitions. Assume that, unless settlement has occurred, each consecutive block of  $b \geq 1$  global rounds grants at least one baseline local opportunity, starting with rounds 1, ...,  $b$ . Foreign deposits can be ignored. Then every admissible play settles by round  $bm$ .

**Proof.** If the play is still unsettled, after  $j$  complete blocks the prover has received at least  $j$  opportunities. Baseline preservation reproduces the first  $j$  baseline transitions. At the end of block  $m$  its baseline certificate has been obtained, unless a different settlement occurred earlier.

## 81. LF-17.2. Eventual service gives eventual settlement without a uniform round bound

LF-17.2 | Src: V49 §17 | Par: {} | f:0 | Hyp: Baseline-preserving emulation hypothesis | Stat: P | Use: Separates per-play termination from a shared worst-case scheduling guarantee.

**Statement.** If an unsettled play grants infinitely many baseline opportunities to a prover whose baseline proof takes finitely many local steps, that play eventually settles. Mere eventual-service fairness does not provide one uniform finite global-round bound across all fair plays.

**Proof.** The required  $m$ th opportunity occurs at some finite round on each such play, yielding the baseline proof. For lack of a uniform bound, use a one-step baseline prover and let the scheduler delay its first service for any chosen finite number  $L$  of rounds, then service it. Every such schedule is fair, but settlement times are unbounded as  $L$  varies.

## 82. LF-A3.1. Certificate-preserved formula properties survive cooperative BANK birth

LF-A3.1 | Src: V48 Int. B §§1–3 | Par: {Source Theorem 3.2} | f:1 | Hyp: source hypotheses | Stat: P | Use: Moves invariant reasoning across channels while keeping proof obligations local.

**Statement.** Every object-level formula deposited at finite BANK birth time has any property  $P$  that holds on  $B_0$  and is preserved by all primitive rules inside accepted expanded deposit certificates.

**Proof.** Theorem 3.2 uses strong induction on BANK birth time plus finite induction through each certificate DAG.

### 83. LF-C10.8. Fixed-degree Polynomial Calculus search is an algorithmic rather than merely existential statement

LF-C10.8 | Src: Core I §10 | Par: {Source Theorem 10.47} | f:1 | Hyp: source hypotheses | Stat: P | Use: Separates bounded structural complexity from unrestricted proof search.

**Statement.** For each fixed degree bound  $d$ , the source theorem supplies a procedure that finds bounded-degree refutations when they exist; existence inside that fixed degree class is therefore accompanied by effective search.

**Proof.** Theorem 10.47 is explicitly a findability theorem at fixed degree. No claim is made when the required degree grows outside its hypotheses.

### 84. LF-SA16.7. Level descends to the admissible quotient

LF-SA16.7 | Src: Core II §§16–17 | Par: {Source Corollary 16.19} | f:1 | Hyp: source hypotheses | Stat: P | Use: Supports degree-level cataloguing of formal self-awareness.

**Statement.** Mutually admissibly equivalent ATPs have equal level, so  $[M] \rightarrow \text{lev}(M)$  is well defined on quotient classes and order preserving.

**Proof.** Corollary 16.19 states both well-definedness and monotonicity on classes.

### 85. LF-24.2. Complete control equates intrinsic and forced horizons

LF-24.2 | Src: V49 §24 | Par: {Source shortest-path/forced-horizon proposition} | f:1 | Hyp: source hypotheses | Stat: P | Use: Identifies a precise case in which two levels of the hierarchy coincide.

**Statement.** In a graph whose every nonterminal choice is controller-owned,  $H_{\text{force}}(s, W) = \text{dist}(s, W)$ .

**Proof.** A finite shortest path can be followed by the controller, providing a forcing strategy of that length. The source inequality provides the reverse bound. If the target is unreachable, neither a path nor a forcing strategy exists.

### 86. LF-11.1. A one-step Bellman equation for proof distance

LF-11.1 | Src: V49 §11 | Par: {Source proof-horizon/distance theorem} | f:1 | Hyp: source hypotheses | Stat: P | Use: Describes what a perfect one-step hint would have to optimize; it does not provide an efficient algorithm for evaluating  $h$ .

**Statement.** Let  $h(S) = \text{dist}(S, \text{WT})$  in the unit-cost logical derivation graph. If  $S \in \text{WT}$ , then  $h(S) = 0$ . Otherwise  $h(S) = 1 + \min U \in \text{Succ}(S) h(U)$ , where  $\min \emptyset = \infty$  and  $1 + \infty = \infty$ .

**Proof.** Every nonzero-length target path has a first successor and a suffix. Conversely a target path from a successor can be prefixed by its incoming edge. Taking minimum lengths in these two directions proves equality.

### 87. LF-A2.2. Conservative BANK growth leaves consequence closure unchanged

LF-A2.2 | Src: V48 Int. B §§1–3 | Par: {Source Theorem 2.2} | f:1 | Hyp: source hypotheses | Stat: P | Use: Formalizes BANK as road-building rather than new axiomatization.

**Statement.** If every accepted object deposit is derivable from the immediately preceding object BANK, then  $\text{ConF}(B_{k\text{obj}}) = \text{ConF}(B_0)$  for every stage  $k$ .

**Proof.** Theorem 2.2 proves both inclusion of every deposited formula in the original closure and equality of closures by monotonicity and idempotence.

### 88. LF-14.2. Finite forced horizon provides a decreasing rank

LF-14.2 | Src: V49 §14 | Par: {Source forced-region construction} | f:1 | Hyp: source hypotheses | Stat: P | Use: Turns the backward construction into a finite progress certificate for a forcing strategy.

**Statement.** If  $s \in W$  has finite forced horizon  $r$ , a controller-owned  $s$  has a successor of horizon at most  $r - 1$ , while every successor of an environment-owned  $s$  has horizon at most  $r - 1$ .

**Proof.** A nonterminal state first enters  $A_r$  through the existential predecessor clause for controller states or the universal nonempty-successor clause for environment states. Those clauses refer to  $A_{r-1}$ .

### 89. LF-SA13.2. Changing the tag spelling does not change the represented machine

LF-SA13.2 | Src: Core II §§13–15 | Par: {Source Proposition 13.4} | f:1 | Hyp: source hypotheses | Stat: P | Use: Prevents notation from being mistaken for a substantive change in self-reference.

**Statement.** If two distinguished tags name the same ATP, uniform renaming between them preserves the underlying tagged-object relation.

**Proof.** This is the renaming invariance of Proposition 13.4.

## 90. LF-I02.1. Structural induction reduces a global WFF property to grammar clauses

LF-I02.1 | Src: V46.1 induction supplement | Par: {Source Supplement Theorem 2} | f:1 | Hyp: source hypotheses | Stat: P | Use: Makes explicit when induction on syntax is justified and when a bare arbitrary WFF set is insufficient.

**Statement.** Under the least-generated grammar hypotheses of Supplement Theorem 2, a property holding on every base formation and preserved by every formation constructor holds for every WFF.

**Proof.** This is the structural induction theorem applied to the grammar's generation tree.

## 91. LF-04.1. Observation spectra descend to degrees

LF-04.1 | Src: V49 §4 | Par: {Source dynamic-monotonicity theorem} | f:1 | Hyp: source hypotheses | Stat: P | Use: Makes observation spectra usable as degree-level database keys, although equality of spectra is not claimed to classify all dynamic degrees.

**Statement.** If  $M \sim_{\text{clk}} N$ , then  $\text{Obs}(M) = \text{Obs}(N)$ . Hence  $[M] \rightarrow \text{Obs}(M)$  is a well-defined monotone function on clocked degrees.

**Proof.** The simulations in the two directions give both inclusions by the source theorem. Monotonicity on representatives then descends using equality on equivalence classes.

## Band C - intermediate structural results, specialized transfers, and operational theorems

Entries 92-137 of 182 in the strength-ordered repository. These entries provide useful intermediate structure, quantitative bounds, or specialized consequences.

## 92. LF-02.1. No finite chain cover

LF-02.1 | Src: V49 §2 | Par: {Source infinite-antichain theorem} | f:1 | Hyp: source hypotheses | Stat: P | Use: Rules out organizing all such machines into finitely many linear ladders of simulation strength.

**Statement.** The clocked-degree poset, even restricted to the finitely describable machines appearing in the source antichain, cannot be covered by finitely many chains.

**Proof.** Let  $A$  be the countably infinite antichain. Each chain meets  $A$  in at most one element. A union of  $m$  chains therefore covers at most  $m$  elements of  $A$ , and cannot cover  $A$ .

## 93. LF-05.2. Effective degrees have no finite chain cover

LF-05.2 | Src: V49 §5 | Par: {Source effective-survival theorem} | f:1 | Hyp: source hypotheses | Stat: P | Use: Shows that restricting to executable witnesses does not restore a finite linear classification.

**Statement.** The effective degree poset is not a well-quasi-order and cannot be covered by finitely many chains.

**Proof.** The source theorem supplies a countably infinite effective antichain. Its enumeration violates well-quasi-ordering, and each chain covers at most one of its elements.

## 94. LF-SA13.3. A proved self-description creates a nonempty self-model

LF-SA13.3 | Src: Core II §§13–15 | Par: {Source Proposition 13.8} | f:1 | Hyp: source hypotheses | Stat: P | Use: Gives the minimal bridge from proof to formal self-model content.

**Statement.** If the machine proves the designated ATP statement about its own tag, its self-model is nonempty.

**Proof.** The proved self-descriptive formula is itself an element witnessing nonemptiness, as recorded in Proposition 13.8.

## 95. LF-I06.2. A nonincreasing rank is an absolute reachability obstruction above the initial maximum

LF-I06.2 | Src: V46.1 induction supplement | Par: {Source Supplement Proposition 17} | f:1 | Hyp: source hypotheses | Stat: P | Use: Turns a potential function into a simple impossibility certificate.

**Statement.** If every inference satisfies  $r(\beta) \leq \max_i r(\alpha_i)$ , then no derived formula can have rank greater than the maximum rank already present in the input.

**Proof.** Proposition 17 says the rank maximum is preserved as an invariant; hence values above it are unreachable.

## 96. LF-I04.2. Rule-driven ATP soundness is a special case of formula-invariant induction

LF-I04.2 | Src: V46.1 induction supplement | Par: {Source Supplement Corollary 8} | f:1 | Hyp: source hypotheses | Stat: P | Use: Shows that proof-search soundness can be packaged as an invariant rather than reproved per stage.

**Statement.** For a rule-driven ATP beginning from  $\Gamma$ , every formula produced is provable from  $\Gamma$ .

**Proof.** Take  $P(\psi)$  to mean  $\Gamma \vdash_F \psi$  in the invariant method. Premises satisfy it and each legal inference preserves it; Corollary 8 records the resulting soundness claim.

## 97. LF-C02.1. Rule-driven states contain no unproved formulas

LF-C02.1 | Src: Core I §§1–2 | Par: {Source Proposition 2.15} | f:1 | Hyp: source hypotheses | Stat: P | Use: Separates sound search failure from the stronger claim of unprovability.

**Statement.** Under the hypotheses of Proposition 2.15, if  $\varphi \in / \text{ConF}(\Gamma)$  then  $\varphi \in / C(\Gamma, m)$  for every stage  $m$ .

**Proof.** Proposition 2.15 gives  $C(\Gamma, m) \subseteq \text{ConF}(\Gamma) \cap Y$ . Take the contrapositive of membership.

## 98. LF-IIT22.1. Mechanism properties and self-theory properties can come apart

LF-IIT22.1 | Src: Core II §22 | Par: {Corrected Proposition 22.4} | f:1 | Hyp: source hypotheses | Stat: P | Use: Prevents a theorem set from being treated as a complete description of mechanism structure.

**Statement.** Within the corrected scope, there is a property of the full presented mechanism that is not determined solely by the self-theory.

**Proof.** Corrected Proposition 22.4 supplies the separation example at the appropriate presentation level.

## 99. LF-SA16.6. A strict level inequality is a non-simulation certificate

LF-SA16.6 | Src: Core II §§16–17 | Par: {LF-SA16.5} | f:1 | Hyp: source hypotheses | Stat: P | Use: Provides an inexpensive obstruction to one direction of the self-awareness preorder.

**Statement.** If  $\text{lev}(M) > \text{lev}(N)$ , then  $M \not\leq_{\text{adm}} \text{SA } N$ .

**Proof.** Otherwise LF-SA16.5 would imply  $\text{lev}(M) \leq \text{lev}(N)$ , a contradiction.

## 100. LF-13.2. Reachable endpoints cannot outnumber histories

LF-13.2 | Src: V49 §13 | Par: {} | f:0 | Hyp: stated conditions + declared background | Stat: P | Use: Quantifies the distinction between proof histories and states without assuming that their counts coincide.

**Statement.** If  $M_n(S_0)$  is finite, the number of distinct endpoints of length- $n$  histories is at most  $M_n(S_0)$ . Equality holds exactly when no two such histories have the same endpoint.

**Proof.** The endpoint map is a surjection from the finite set of histories onto its image. Equal cardinalities occur exactly when that map is injective.

## 101. LF-SA15.3. Provability of a reflection formula places it in the self-theory

LF-SA15.3 | Src: Core II §§13–15 | Par: {Source Corollary 15.13} | f:1 | Hyp: source hypotheses | Stat: P | Use: Connects reflection achievement to the machine's designated self-descriptive theorem set.

**Statement.** If  $M \vdash \theta_k$ , then  $\theta_k \in \text{Self}(M)$ .

**Proof.** Corollary 15.13 records this because each  $\theta_k$  is self-descriptive and proved by  $M$ .

## 102. LF-SA15.4. Finite levels are cumulative under provability soundness

LF-SA15.4 | Src: Core II §§13–15 | Par: {Source Theorem 15.22} | f:1 | Hyp: source hypotheses | Stat: P | Use: Makes the finite hierarchy nested once its necessary hypothesis is stated.

**Statement.** Under the theorem's soundness hypotheses, Level  $n$  implies every lower positive finite level.

**Proof.** Theorem 15.22 proves cumulativity by unpacking the reflection sequence and the relevant soundness condition.

## 103. LF-SA16.5. Level is monotone under admissible comparison

LF-SA16.5 | Src: Core II §§16–17 | Par: {Source Theorem 16.18} | f:1 | Hyp: source hypotheses | Stat: P | Use: Turns the level function into an order invariant rather than a presentation artifact.

**Statement.** If  $M \leq_{\text{adm}} \text{SA } N$ , then  $\text{lev}(M) \leq \text{lev}(N)$ .

**Proof.** This is Theorem 16.18, based on preservation of reflection formulas under the admissible translation.

## 104. LF-A4.2. Later refinement cannot move first settlement later

LF-A4.2 | Src: V48 Int. B §§4–5 | Par: {Source Proposition 4.2} | f:1 | Hyp: source hypotheses | Stat: P | Use: Allows dependence to be refined by a proof or refutation without rewriting history.

**Statement.** If first settlement occurs at stage  $h$ , later accepted settlement records may enlarge  $\Sigma_k$  but the stored first-settlement time remains  $h$ .

**Proof.** Proposition 4.2 uses the absorbing first-time field and append-only settlement set.

## 105. LF-A4.3. A proof arriving after dependence refines status without changing historical first settlement

LF-A4.3 | Src: V48 Int. B §§4–5 | Par: {LF-A4.2} | f:1 | Hyp: source hypotheses | Stat: P | Use: Separates earliest certified information from later, more specific certified information.

**Statement.** If  $D$  settles first at stage  $h$  and  $P$  later supplies a valid proof at stage  $k > h$ , then both records may be present at stage  $k$  while  $\tau_{\text{Aset}} = h$ .

**Proof.** LF-A4.2 permits  $\Sigma_h \subseteq \Sigma_k$  while keeping the first-settlement time absorbing.

### 106. LF-S21.1. A purely formal reflection-closure machine realizes level omega

LF-S21.1 | Src: Core II §§20–21 | Par: {Source Theorem 21.1} | f:1 | Hyp: source hypotheses | Stat: P | Use: Gives a non-human formal witness for the top finite-reflection level.

**Statement.** There exists a tagged ATP R in the source construction with level  $\omega$  that derives the full finite reflection sequence.

**Proof.** Theorem 21.1 supplies the machine and its properties.

### 107. LF-TM.B2. The exhaustive configuration enumerator is sound and complete

LF-TM.B2 | Src: Corrected App. B | Par: {Source Proposition B.8} | f:1 | Hyp: source hypotheses | Stat: P | Use: Provides an explicit computational example of the general rule-enumeration architecture.

**Statement.** The SIC ET is rule-driven and deductively effective over the configuration formal system.

**Proof.** This is Proposition B.8.

### 108. LF-A6.1. A nonstarved baseline-preserving prover cannot lose the stand-alone logical-round deadline

LF-A6.1 | Src: V48 Int. B §§6–7 | Par: {Source Theorem 6.3} | f:1 | Hyp: source hypotheses | Stat: P | Use: Provides a no-regret guarantee under a precisely stated logical-round resource convention.

**Statement.** If the stand-alone prover proves T in m local transitions, then under baseline preservation and unit-rate nonstarvation the AMLD either settles earlier through another role or reproduces the proof by global round m.

**Proof.** Theorem 6.3 gives P at least m opportunities by round m and lets it ignore foreign deposits and mimic the stand-alone path.

### 109. LF-08.1. Finite support of a derived theorem

LF-08.1 | Src: V49 §8 | Par: {} | f:0 | Hyp: declared definition/background | Stat: P | Use: Localizes an infinite premise pool to the finite support of an individual certificate.

**Statement.** If  $B \vdash F T$ , then some finite  $B^* \subseteq B$  satisfies  $B^* \vdash F T$ .

**Proof.** A finite proof has only finitely many premise occurrences. Collect those members of B used in the proof into  $B^*$ . The same proof is valid from  $B^*$ .

### 110. LF-04.2. Finite-pattern obstruction to simulation

LF-04.2 | Src: V49 §4 | Par: {Source dynamic-monotonicity theorem} | f:1 | Hyp: source hypotheses | Stat: P | Use: Gives a targeted theorem-search objective: look for a synchronized pattern realized on only one side.

**Statement.** If a finite pattern lies in  $\text{Obs}(M) \setminus \text{Obs}(N)$ , then  $M \not\leq_{\text{clk}} N$ . If there is also a pattern in  $\text{Obs}(N) \setminus \text{Obs}(M)$ , the two machines are incomparable.

**Proof.** A simulation would force the inclusion contradicted by the first pattern. Apply the same argument in reverse for incomparability.

### 111. LF-I04.1. Formula invariants persist at every stage of a rule-driven ATP

LF-I04.1 | Src: V46.1 induction supplement | Par: {Source Supplement Theorem 7} | f:1 | Hyp: source hypotheses | Stat: P | Use: Turns local rule checks into a run-wide search invariant.

**Statement.** Under the hypotheses of ATP formula-invariant induction, every formula appearing at any stage has the invariant property.

**Proof.** The theorem inducts over stages: old formulas retain the property and newly admitted formulas inherit it from rule-preserving premises.

### 112. LF-A8.1. Cumulative certificate depth upper-bounds attainable rank

LF-A8.1 | Src: V48 Int. B §§8–11 | Par: {Source Theorem 8.1} | f:1 | Hyp: source hypotheses | Stat: P | Use: Converts a local inference constraint into a global cooperative growth bound.

**Statement.** Under the theorem's local rank-increment bound d, every formula in  $B_{\text{obj}}$  has rank at most  $r_0 + dQ_n$ .

**Proof.** Theorem 8.1 accumulates at most  $dq_i$  additional rank across each accepted certificate segment and sums those segment bounds.

### 113. LF-C05.1. A union of sound branches is sound

LF-C05.1 | Src: Core I §§5–6 | Par: {Source Theorem 5.6} | f:1 | Hyp: source hypotheses | Stat: P | Use: Allows nondeterministic exploration to pool results without weakening correctness.

**Statement.** If every branch state generated under the hypotheses of the branch-soundness theorem contains only consequences of  $\Gamma$ , then the union of all branch states also contains only consequences of  $\Gamma$ .

**Proof.** Each element of the union lies in some branch state, which is sound by Theorem 5.6.

### 114. LF-I08.1. Internal necessitation propagates finite reflection one level at a time

LF-I08.1 | Src: V46.1 induction supplement | Par: {Source Supplement Proposition 19} | f:1 | Hyp: source hypotheses | Stat: P | Use: Separates finite reflection induction from any transfinite limit construction.

**Statement.** If a tagged machine proves the base reflection formula and has the source internal-necessitation property, then it proves every finite reflection iterate.

**Proof.** Proposition 19 performs ordinary induction on the reflection index, using internal necessitation for the successor step.

### 115. LF-06.1. Infinite width does not exclude a greatest element

LF-06.1 | Src: V49 §6 | Par: {} | f:0 | Hyp: declared construction/background | Stat: P | Use: Blocks an invalid inference from the source antichain theorem.

**Statement.** There is a countable poset with a countably infinite antichain and a greatest element.

**Proof.** Take distinct elements  $a_0, a_1, \dots$  and a new element  $T$ . Declare  $a_i \leq T$  for all  $i$ , retain reflexivity, and introduce no comparisons between distinct  $a_i$ . This is a poset with the stated properties.

### 116. LF-SA13.4. Self-theory contains only the machine's theorems

LF-SA13.4 | Src: Core II §§13–15 | Par: {Source Proposition 13.10} | f:1 | Hyp: source hypotheses | Stat: P | Use: Places the self-theory firmly inside ordinary theoremhood.

**Statement.** For every tagged ATP  $M$ ,  $\text{Self}(M)$  is a subset of the formulas provable by  $M$ .

**Proof.** This is Proposition 13.10 by the definition of self-theory as the proved self-descriptive fragment.

### 117. LF-20.2. Safe ranking can nevertheless miss a reachable target forever

LF-20.2 | Src: V49 §20 | Par: {} | f:0 | Hyp: declared definition/background | Stat: P | Use: Shows why an independent fairness condition is needed when the aim is eventual discovery.

**Statement.** There is a sound legal-action policy with infinite hitting time for a reachable target.

**Proof.** Let an operational state have a legal self-loop and a legal edge to an accepted target. The policy that always selects the self-loop uses only legal actions and never accepts a false theorem, but never visits the target.

### 118. LF-27.1. A consistent potential lower-bounds target distance

LF-27.1 | Src: V49 §27 | Par: {} | f:0 | Hyp: stated conditions + declared background | Stat: P | Use: Supplies one sufficient bridge for the source's admissible-heuristic question.

**Statement.** Let  $\ell: S \rightarrow [0, \infty)$  satisfy  $\ell(w) = 0$  for  $w \in W$  and  $\ell(s) \leq 1 + \ell(t)$  on every legal edge  $s \rightarrow t$ . Then  $\ell(s) \leq \text{dist}(s, W)$ .

**Proof.** Along any length- $n$  path to a target, repeated application of the edge inequality gives  $\ell(s) \leq n + \ell(w) = n$ . Minimize over target paths. When none exists the claimed upper bound is infinite.

### 119. LF-A7.2. A wall-clock win can coexist with greater total work

LF-A7.2 | Src: V48 Int. B §§6–7 | Par: {Source Proposition 6.5 and Proposition 7.1} | f:2 | Hyp: source hypotheses | Stat: P | Use: Provides a concrete two-metric caution for evaluating cooperative ATPs.

**Statement.** Under replicated resources the AMLD may settle no later in wall-clock time than the stand-alone prover while still performing more aggregate work across its roles.

**Proof.** Proposition 6.5 controls wall-clock time under replication. Proposition 7.1 explicitly allows excess work by losing roles. The two metrics therefore need not move together.

### 120. LF-A4.1. First settlement is the minimum of the five role times

LF-A4.1 | Src: V48 Int. B §§4–5 | Par: {Source Proposition 4.1} | f:1 | Hyp: source hypotheses | Stat: P | Use: Gives an exact aggregation law for the multi-role architecture.

**Statement.** In the extended natural numbers,  $\tau_{\text{Aset}}(T) = \min\{\tau_{\text{AP}}, \tau_{\text{AR}}, \tau_{\text{AD}}, \tau_{\text{AI}}, \tau_{\text{AC}}\}$ .

**Proof.** Proposition 4.1 unfolds the definition of first acceptance among the five designated certificate classes.

### 121. LF-C07.3. Finite-workload compilation does not imply an asymptotic speedup family

LF-C07.3 | Src: Core I §§7–9 | Par: {Source Corollary 7.14} | f:1 | Hyp: source hypotheses | Stat: P | Use: Prevents a benchmark optimization from being promoted into a complexity-class claim.

**Statement.** A speedup obtained by compiling finitely many targets does not by itself prove an asymptotic proof complexity separation on an infinite target family.

**Proof.** Corollary 7.14 states exactly this scope boundary: finite designated-target compilation is not an asymptotic theorem.

### 122. LF-A2.3. Conservative cooperation cannot change provability status of $T$ , $T$ -minus, or contradiction

LF-A2.3 | Src: V48 Int. B §§1–3 | Par: {Source Corollary 2.3} | f:1 | Hyp: source hypotheses | Stat: P | Use: Separates improved access from changed logical status.

**Statement.** Under conservative object-BANK growth, the truth values of  $B_k \vdash T$ ,  $B_k \vdash T \neg$ , and  $B_k \vdash \perp$  are independent of  $k$ .

**Proof.** Corollary 2.3 follows because the consequence closure is constant.

### 123. LF-SA15.6. Sound internal necessitation creates a zero-or-omega dichotomy

LF-SA15.6 | Src: Core II §§13–15 | Par: {Source Proposition 15.29} | f:1 | Hyp: source hypotheses | Stat: P | Use: Identifies a structural condition that collapses the finite middle of the hierarchy.

**Statement.** For a provability-sound ATP with internal necessitation, the level is either 0 or  $\omega$ ; no positive finite intermediate level occurs.

**Proof.** This is Proposition 15.29: once Level 1 is present, internal necessitation drives all finite reflection; otherwise the level remains zero.

### 124. LF-A5.1. Object-BANK enlargement cannot increase P/R/C shortest explicitpremise horizons

LF-A5.1 | Src: V48 Int. B §§4–5 | Par: {Source Theorem 5.1} | f:1 | Hyp: source hypotheses | Stat: P | Use: Formalizes the narrow sense in which shared verified lemmas can only improve available proof routes.

**Statement.**  $\text{obj For } J \in \{P, R, C\}, hJ(k+1) \leq hJ(k) \text{ whenever } Bk \text{obj} \subseteq Bk+1.$

**Proof.** Every old proof remains a legal proof from the enlarged premise set, so the new minimum cannot be larger.

### 125. LF-23.1. Distance and realized hitting time distribute over finite target unions

LF-23.1 | Src: V49 §23 | Par: {} | f:0 | Hyp: declared definition/background | Stat: P | Use: Explains the first-settlement minimum while retaining the typing of the separate certificate regions.

**Statement.** For  $W = \bigcup_{i=1}^r W_i$  with  $r \geq 1$ ,  $\text{Sr dist}(s, W) = \min \text{dist}(s, W_i)$ ,  $i \text{ H}\pi_{\text{hit}}(W) = \min \text{H}\pi_{\text{hit}}(W_i)$ .  $i$  Both equalities allow infinite values.

**Proof.** A path, or a time on a fixed trajectory, reaches the union exactly when it reaches at least one constituent region. Taking the earliest such length or time gives the minimum.

### 126. LF-SA17.2. The Tegmark condition occurs at every positive advertised level

LF-SA17.2 | Src: Core II §§16–17 | Par: {Source Proposition 17.9} | f:1 | Hyp: source hypotheses | Stat: P | Use: Shows that the Tegmark condition does not force one unique reflection level.

**Statement.** For every positive finite level and for  $\omega$  in the source range, there is a tagged ATP satisfying the Tegmark condition at that level.

**Proof.** Proposition 17.9 supplies the witnesses.

### 127. LF-21.1. Sharing repeated lemma uses avoids repeated expansion

LF-21.1 | Src: V49 §21 | Par: {} | f:0 | Hyp: explicit certificate hypotheses | Stat: P | Use: Quantifies proof sharing under a declared DAG cost model.

**Statement.** Suppose a lemma  $\lambda$  has a fixed acyclic certificate from B with  $q$  primitive inference nodes. Suppose a later certificate for T from  $B \cup \{\lambda\}$  has  $r$  primitive inference nodes and any finite number of uses of  $\lambda$ . In a reusable certificate-DAG representation there is a certificate of T from B with at most  $q + r$  inference nodes.

**Proof.** Insert the lemma's DAG once before the later certificate and redirect every premise reference to  $\lambda$  to its conclusion node. Retain the later  $r$  nodes and all their dependencies. The resulting graph is acyclic because the lemma's derivation depends only on B, and every redirected edge points from that earlier derivation into the later one.

### 128. LF-23.2. Forcing a settlement union can be easier than forcing any fixed role

LF-23.2 | Src: V49 §23 | Par: {} | f:0 | Hyp: declared definition/background | Stat: P | Use: A strategic consequence specific to multi-channel settlement: robust first settlement need not identify in advance which role will settle.

**Statement.** Always  $\text{Hforce}(s, \bigcup W_i) \leq \min_i \text{Hforce}(s, W_i)$ . Equality need not hold: the left side can be one while every term on the right is infinite. S

**Proof.** Any strategy forcing one region forces the union. For strictness, let the environment choose in one step between two terminal states  $w_1$  and  $w_2$ , and take  $W_i = \{w_i\}$ . Every choice reaches the union in one step. For either fixed target  $W_i$ , the environment can choose the other terminal and avoid it forever.

### 129. LF-C04.4. Recoded SICs occupy the same synchronous and clocked degree

LF-C04.4 | Src: Core I §§3–4 | Par: {Source Theorem 4.12} | f:1 | Hyp: source hypotheses | Stat: P | Use: Separates representational identity from genuinely different operational power.

**Statement.** For a SIC and its invertibly recoded copy, both synchronous and clocked mutual simulations hold.

**Proof.** Theorem 4.12 supplies the two simulation directions using the recoding and its inverse.

### 130. LF-C10.5. Polynomial Calculus inherits polynomial translations of resolution proofs

LF-C10.5 | Src: Core I §10 | Par: {Source Theorem 10.43} | f:1 | Hyp: source hypotheses | Stat: P | Use: Provides a one-way proof-translation guarantee while allowing the reverse direction to be harder.

**Statement.** Every resolution proof in the source setting has a Polynomial Calculus proof with polynomially related size.

**Proof.** Theorem 10.43 states that Polynomial Calculus p-simulates resolution. Unpack the definition of p-simulation.

### 131. LF-05.1. Ordinary non-simulation survives effectiveness restrictions

LF-05.1 | Src: V49 §5 | Par: {} | f:0 | Hyp: declared definition/background | Stat: P | Use: Transfers negative results without requiring a separate computability obstruction.

**Statement.** If  $M/\leq_{\text{clk}} N$  for unrestricted clocked witnesses, there is no effective clocked simulation from  $M$  to  $N$ .

**Proof.** An effective witness is also an unrestricted witness. Existence of the former would contradict absence of the latter.

### 132. LF-26.1. A bounded unsuccessful search trace does not certify unreachability

LF-26.1 | Src: V49 §26 | Par: {} | f:0 | Hyp: declared construction/background | Stat: P | Use: Shows the information missing from a timeout trace alone.

**Statement.** For every  $n \geq 0$ , there are two rooted transition systems with identical observed nonterminal prefixes through  $n$  steps, one with a target at step  $n + 1$  and one with no reachable target.

**Proof.** Use a deterministic chain  $s_0 \rightarrow \dots \rightarrow s_n$  with no target labels. In one extension add  $s_n \rightarrow w$  with  $w$  a target. In the other add an edge from  $s_n$  to a new non-target state with a self-loop. The prefixes and observations through step  $n$  coincide, but their reachability answers differ.

### 133. LF-P20.4. Level omega does not exhaust Picard's self-theory

LF-P20.4 | Src: Core II §§20–21 | Par: {Source Corollary 20.17} | f:1 | Hyp: source hypotheses | Stat: P | Use: Shows that the level invariant records only one structured slice of self-descriptive theoremhood.

**Statement.** Under the Picard idealization, the self-theory strictly extends the finite reflection tower rather than being equal to it.

**Proof.** Corollary 20.17 states the strict extension.

### 134. LF-C04.5. Pure invertible recoding cannot by itself create a transaction advantage

LF-C04.5 | Src: Core I §§3–4 | Par: {Source Theorem 4.34} | f:1 | Hyp: source hypotheses | Stat: P | Use: Guards theorem-fishing experiments against counting relabeling as computational progress.

**Statement.** Under the hypotheses of the no-free-recoding-speedup theorem, any claimed transaction reduction must use more than the invertible recoding alone.

**Proof.** Theorem 4.34 excludes a speedup whose only change is pure invertible recoding. Therefore a genuine improvement needs a changed search policy, rules, compilation, resource model, or another non-recoding ingredient.

### 135. LF-13.1. Uniform branching bounds history growth

LF-13.1 | Src: V49 §13 | Par: {Source exact derivation-history recursion} | f:1 | Hyp: source hypotheses | Stat: P | Use: Supplies a finite enumeration bound under an explicit branching bound; finitary inference alone does not imply finite branching.

**Statement.** If every state reachable within the first  $n$  steps has at most  $b < \infty$  legal labeled actions, then  $M_n(S_0) \leq b^n$ , using  $b_0 = 1$  also for  $b = 0$ .

**Proof.** At depth zero there is one empty history. Each length- $j$  history has at most  $b$  extensions, so induction gives at most  $b^j$  histories at depth  $j \leq n$ .

### 136. LF-A8.4. Causal BANK depth never exceeds the cumulative depth budget at deposit time

LF-A8.4 | Src: V48 Int. B §§8–11 | Par: {Source Theorem 8.3} | f:1 | Hyp: source hypotheses | Stat: P | Use: Orders the two AMLD potential measures and explains why the causal one can be sharper.

**Statement.** If  $\lambda$  is deposited by update  $n$ , then  $\delta(\lambda) \leq Q_n$ .

**Proof.** Theorem 8.3 observes that each dependency segment used by the target is bounded by its deposit's  $q_i$ , while  $Q_n$  sums all segment budgets up to  $n$ .

### 137. LF-P20.1. Picard attains level omega under the formal idealization

LF-P20.1 | Src: Core II §§20–21 | Par: {Source Theorem 20.4} | f:1 | Hyp: source hypotheses | Stat: P | Use: Provides a named level- $\omega$  witness without turning the result into an empirical claim about a human mind.

**Statement.** Under Definition 20.1's Picard idealization,  $\text{lev}(\text{Picard}) = \omega$ .

**Proof.** This is the Picard theorem of Section 20. The conclusion is about the formal idealization under its stated premises.

## Band D - elementary checks, boundary results, local counterexamples, and scope-control consequences

Entries 138–182 of 182 in the strength-ordered repository. These entries are retained because boundary checks and negative scope controls are part of an auditable theorem repository, even when the mathematics is elementary.

### 138. LF-SA16.1. Unrestricted translations are too weak to support a discriminating order

LF-SA16.1 | Src: Core II §§16–17 | Par: {Source Proposition 16.3} | f:1 | Hyp: source hypotheses | Stat: P | Use: Explains why a syntax-respecting translation class is mathematically substantive.

**Statement.** If translations are allowed without the source admissibility restrictions, the proposed comparison collapses in the manner shown by Proposition 16.3.

**Proof.** Proposition 16.3 constructs the collapse, so admissibility is a necessary design response within this framework.

### 139. LF-I04.3. Birth-time induction can prove properties unavailable from stage membership alone

LF-I04.3 | Src: V46.1 induction supplement | Par: {Source Supplement Theorem 10} | f:1 | Hyp: source hypotheses | Stat: P | Use: Provides the ancestor of the later cooperative BANK birth-time method.

**Statement.** If a property is established for inputs and for every formula at the moment it is first born from earlier formulas with that property, then every formula with finite birth time has the property.

**Proof.** Apply strong induction to the natural-number birth time as in Supplement Theorem 10.

### 140. LF-D18.2. The Discovery witness class survives strengthening to any realized positive level

LF-D18.2 | Src: Core II §18 | Par: {Source Proposition 18.7} | f:1 | Hyp: source hypotheses | Stat: P | Use: Shows the existential content is not tied to the weakest positive level.

**Statement.** Under the source strengthening parameterized by a positive finite level or  $\omega$ , the existence claim remains inhabited as stated in Proposition 18.7.

**Proof.** The proposition reuses the hierarchy witnesses to meet the stronger level requirement.

### 141. LF-P20.2. The Picard idealization is a nondegenerate formal witness

LF-P20.2 | Src: Core II §§20–21 | Par: {Source Corollary 20.7} | f:1 | Hyp: source hypotheses | Stat: P | Use: Distinguishes nondegeneracy of a formal witness from mere Level- $\omega$  attainment.

**Statement.** Under the same idealization and the corrected nondegeneracy criterion, Picard is a nondegenerate witness in the source sense.

**Proof.** Corollary 20.7 records the conclusion from the Picard theorem plus the criterion.

### 142. LF-SA17.1. The Tegmark condition is upward closed under admissible comparison

LF-SA17.1 | Src: Core II §§16–17 | Par: {Source Proposition 17.6} | f:1 | Hyp: source hypotheses | Stat: P | Use: Makes the condition compatible with the admissible order rather than isolated to single presentations.

**Statement.** If  $M$  satisfies the source Tegmark condition and  $M \leq_{\text{adm}} \text{SA } N$ , then  $N$  also satisfies it.

**Proof.** This is the upward-closure proposition of Section 17.

### 143. LF-24.1. Intrinsic distance lower-bounds any realized successful route

LF-24.1 | Src: V49 §24 | Par: {} | f:0 | Hyp: declared definition/background | Stat: P | Use: Valid only when the two quantities refer to the same graph and cost unit; primitive proof steps and ATP expansions are not automatically comparable.

**Statement.** If a policy trajectory uses edges of a fixed unit-cost graph and starts at  $s$ , then  $\text{dist}(s, W) \leq H_{\text{thit}}(W)$ .

**Proof.** If the policy hits at time  $n$ , its prefix is an  $n$ -edge path to  $W$ , so shortest distance is at most  $n$ . If it never hits, the right side is infinite.

### 144. LF-I07.1. Halting persistence implies permanent state freezing under the SIC axioms

LF-I07.1 | Src: V46.1 induction supplement | Par: {Source Supplement Proposition 18} | f:1 | Hyp: source hypotheses | Stat: P | Use: Supports stable settlement records after target attainment.

**Statement.** Once  $H(\Gamma, h) = 1$ , every later stage remains halted and has the same frozen state required by the SIC definition.

**Proof.** Proposition 18 propagates the halting predicate forward; the SIC freezing axiom then propagates the state equality stage by stage.

### 145. LF-IIT22.2. Phi is typed as a mechanism property in the source comparison

LF-IIT22.2 | Src: Core II §22 | Par: {Source Proposition 22.6} | f:1 | Hyp: source hypotheses | Stat: P | Use: Keeps a conditional cross-framework comparison from collapsing distinct mathematical objects.

**Statement.** Under the source's chosen IIT terminology,  $\Phi$  belongs to the mechanism-description side rather than being a property determined by self-theory alone.

**Proof.** Proposition 22.6 states this type distinction.

### 146. LF-TM.B3. Trace SICs are computable for deterministic Turing machines

LF-TM.B3 | Src: Corrected App. B | Par: {Source Proposition B.17} | f:1 | Hyp: source hypotheses | Stat: P | Use: Shows a broad class of SICs are ordinary effective computational objects.

**Statement.** For a deterministic Turing machine with the source effective encoding, the corresponding trace SIC has a computable stage evolution.

**Proof.** Proposition B.17 simulates the finite number of machine steps required to obtain each encoded stage.

### 147. LF-TM.B4. Effective-state computable SICs admit Turing presentations

LF-TM.B4 | Src: Corrected App. B | Par: {Corrected Theorem B.18} | f:1 | Hyp: source hypotheses | Stat: P | Use: Closes the representation gap that would otherwise hide noncomputable state access.

**Statement.** On an effectively presented state domain, the computable SIC notion and the source Turing-machine realization notion coincide as stated by corrected Theorem B.18.

**Proof.** The correction supplies the exact equivalence only after the state-domain effectiveness requirement is imposed.

### 148. LF-C01.3. Premise enlargement cannot lengthen a shortest proof

LF-C01.3 | Src: Core I §§1–2 | Par: {} | f:0 | Hyp: declared definition/background | Stat: P | Use: This monotonicity is the object-level template later reused for BANK road-building.

**Statement.** If  $A \subseteq B \subseteq y$ , then  $\|\varphi\|F(B) \leq \|\varphi\|F(A)$  in  $N \cup \{\infty\}$ .

**Proof.** Every proof from A is also a proof from B. The set over which the minimum is taken for B therefore contains the set for A.

### 149. LF-S21.3. Formal universality is an existence result inside the specified framework

LF-S21.3 | Src: Core II §§20–21 | Par: {Source Theorem 21.4} | f:1 | Hyp: source hypotheses | Stat: P | Use: Records settlement without extending it beyond the formalized content.

**Statement.** The formal content of the source universality conjecture is settled in the affirmative in the precise sense stated by Theorem 21.4.

**Proof.** Theorem 21.4 supplies the formal witness and explicitly scopes the conclusion.

### 150. LF-SA15.5. Internal necessitation plus Level 1 yields Level omega

LF-SA15.5 | Src: Core II §§13–15 | Par: {Source Theorem 15.27} | f:1 | Hyp: source hypotheses | Stat: P | Use: Provides a sufficient mechanism for maximal finite reflection.

**Statement.** A tagged ATP satisfying the hypotheses of Theorem 15.27 proves every finite reflection iterate and therefore attains level  $\omega$ .

**Proof.** Theorem 15.27 iterates internal necessitation through all finite reflection levels.

### 151. LF-SA15.2. Distinct reflection formulas prevent level collapse by syntactic identity

LF-SA15.2 | Src: Core II §§13–15 | Par: {Source Proposition 15.14} | f:1 | Hyp: source hypotheses | Stat: P | Use: Protects the hierarchy from trivial syntactic collapse.

**Statement.** The finite reflection sequence  $\theta_0, \theta_1, \dots$  contains pairwise distinct formulas, so proving a later level cannot be explained by the later formula being literally the same string as an earlier one.

**Proof.** This is the pairwise-distinctness conclusion of Proposition 15.14.

### 152. LF-S21.6. Transfinite reflection levels are pairwise syntactically distinct in the specified calculus

LF-S21.6 | Src: Core II §§20–21 | Par: {Source Proposition 21.14} | f:1 | Hyp: source hypotheses | Stat: P | Use: Provides the syntactic separation needed before asking whether different ordinal levels are semantically or order-theoretically distinct.

**Statement.** For the transfinite sequence considered in Section 21,  $\theta_\alpha \neq \theta_\beta$  whenever  $\alpha \neq \beta < \kappa$ .

**Proof.** This is Proposition 21.14.

### 153. LF-A7.1. AMLD architecture alone gives no universal total-work inequality

LF-A7.1 | Src: V48 Int. B §§6–7 | Par: {Source Proposition 7.1} | f:1 | Hyp: source hypotheses | Stat: P | Use: Separates parallel opportunity from aggregate computational cost.

**Statement.** Neither  $WA \leq W\star$  nor  $WA \geq W\star$  holds on every target merely from using five roles.

**Proof.** Proposition 7.1 gives cases in both directions: losing roles can waste work, while early alternative settlement can terminate before the baseline spends its full work budget.

### 154. LF-C09.3. Computability requires an effective state presentation

LF-C09.3 | Src: Core I §§7–9 | Par: {Corrected Theorem 9.15} | f:1 | Hyp: source hypotheses | Stat: P | Use: Prevents hidden coding assumptions in ATP/Turing comparisons.

**Statement.** A claim that a SIC is simulated by a Turing machine is licensed only after the SIC state domain is effectively presented as required by corrected Theorem 9.15.

**Proof.** The correction makes effective presentation an explicit hypothesis. Dropping it would ask a Turing machine to operate on states without a computable representation.

### 155. LF-P20.6. Finite observed performance does not refute a dispositional level-omega idealization

LF-P20.6 | Src: Core II §§20–21 | Par: {Source Proposition 20.21} | f:1 | Hyp: source hypotheses | Stat: P | Use: Clarifies a competence/performance distinction inside the formal model.

**Statement.** Under the source dispositional reading, having proved only finitely many reflection formulas by a given time is compatible with the formal Level- $\omega$  idealization.

**Proof.** Proposition 20.21 separates finite current performance from the dispositional premise used by the theorem.

### 156. LF-P20.5. The Picard construction is uniform over any system satisfying the same formal premises

LF-P20.5 | Src: Core II §§20–21 | Par: {Source Proposition 20.19} | f:1 | Hyp: source hypotheses | Stat: P | Use: Identifies which part of the argument is structural rather than biographical.

**Statement.** Replacing Picard by an arbitrary tagged ATP  $X$  satisfying the formal premises (P1)–(P4) yields the same corresponding formal conclusion stated in Proposition 20.19.

**Proof.** Proposition 20.19 is the uniformity theorem: the proof uses the premises, not an irreducible proper name.

### 157. LF-SA13.1. Self-representation does not require arithmetic coding

LF-SA13.1 | Src: Core II §§13–15 | Par: {Source Proposition 13.2} | f:1 | Hyp: source hypotheses | Stat: P | Use: Keeps the framework syntactic rather than tying it to a particular numeral coding.

**Statement.** A tagged ATP may have a valid internal representation even when that representation is introduced as a formal name rather than as a Gödel number.

**Proof.** Proposition 13.2 explicitly allows a syntactic name with intended denotation and imposes no arithmetic-encoding requirement.

### 158. LF-S21.2. The Picard premises are jointly formally instantiable

LF-S21.2 | Src: Core II §§20–21 | Par: {Source Corollary 21.2} | f:1 | Hyp: source hypotheses | Stat: P | Use: Separates consistency/instantiability of the formal premise package from empirical attribution.

**Statement.** The source assumptions (P1)–(P4) are jointly satisfiable by a formal construction.

**Proof.** Corollary 21.2 discharges the formal instantiability question using the reflection-closure machine.

### 159. LF-D18.1. Positive formal self-awareness is inhabited

LF-D18.1 | Src: Core II §18 | Par: {Source Proposition 18.2} | f:1 | Hyp: source hypotheses | Stat: P | Use: Separates an existence theorem from broader philosophical interpretations.

**Statement.** The class of tagged ATPs with positive formal self-awareness is nonempty.

**Proof.** Proposition 18.2 supplies explicit witness existence from the earlier constructions.

### 160. LF-S21.4. Competence and finite performance are formally separable

LF-S21.4 | Src: Core II §§20–21 | Par: {Source Theorem 21.6} | f:1 | Hyp: source hypotheses | Stat: P | Use: Prevents a finite run prefix from being identified with the full dispositional theory.

**Statement.** The source formalization admits a system with the relevant competence/dispositional property even when only finite performance is observed at a stage.

**Proof.** Theorem 21.6 is the formal content of the competence–performance gap.

### 161. LF-ML1.1. Changing only the legal-move ranking cannot create a false theorem

LF-ML1.1 | Src: V48 Int. A | Par: {Source Proposition 1.2} | f:1 | Hyp: source hypotheses | Stat: P | Use: Formalizes the trust boundary for learned or heuristic search guidance.

**Statement.** With a fixed sound verifier, fixed formal environment, and unchanged certificate semantics, replacing one legal-move scoring/ranking policy by another cannot make the ATP accept a non-theorem.

**Proof.** Proposition 1.2 separates proposal order from verifier acceptance: only verifier-accepted certificates enter the trusted output.

### 162. LF-SA15.1. Positive Level 1 implies a nonempty self-theory

LF-SA15.1 | Src: Core II §§13–15 | Par: {Source Proposition 15.4} | f:1 | Hyp: source hypotheses | Stat: P | Use: Makes Level 1 a genuinely nonempty formal condition.

**Statement.** Every Level-1 formally self-aware ATP has at least one theorem in its self-theory.

**Proof.** Proposition 15.4 supplies the implication directly.

### 163. LF-31.2. One layer can hide the full range of discovery offsets

LF-31.2 | Src: V49 §31 | Par: {} | f:0 | Hyp: declared construction/background | Stat: P | Use: Shows what ordering can change before asserting any special advantage for Hilbert, lexicographic, learned, or randomized policies.

**Statement.** A finite layer of  $N \geq 1$  candidates with exactly one successful candidate admits an ordering that succeeds first and another that succeeds at position  $N$ , without changing the successful depth.

**Proof.** Place the successful candidate first or last. Both orderings contain the same candidates in the same layer.

#### 164. LF-A9.1. AMLD introduces no fifth foundational induction axiom

LF-A9.1 | Src: V48 Int. B §§8–11 | Par: {Source Proposition 9.1} | f:1 | Hyp: source hypotheses | Stat: P | Use: Places cooperative reasoning inside the existing induction hierarchy rather than advertising a new foundational principle.

**Statement.** Global AMLD induction and cooperative BANK birth-time induction are ordinary inductions on natural-number chronologies, supplemented by finite induction through certificate DAGs.

**Proof.** This is Proposition 9.1.

#### 165. LF-08.2. Zero inference horizon identifies existing premises

LF-08.2 | Src: V49 §8 | Par: {} | f:0 | Hyp: stated conditions + declared background | Stat: P | Use: Fixes a base case used in BANK-horizon calculations.

**Statement.** For finite initial premises  $B$ ,  $\text{HFinf}(B, T) = 0$  if and only if  $T \in B$ .

**Proof.** An existing premise requires no inference. Conversely, a proof with no non-premise inference applications contains only premises and cannot introduce a different target.

#### 166. LF-09.1. Enlarging the target cannot increase distance

LF-09.1 | Src: V49 §9 | Par: {} | f:0 | Hyp: declared definition/background | Stat: P | Use: Separates target enlargement from adding moves or changing the search policy.

**Statement.** For target regions  $W \subseteq W'$  and any state  $s$ ,  $\text{dist}(s, W') \leq \text{dist}(s, W)$ , with  $\infty$  allowed.

**Proof.** Every path ending in  $W$  also ends in  $W'$ . Minimize over the larger set of eligible paths.

#### 167. LF-03.2. A simple sufficient condition for a static meet

LF-03.2 | Src: V49 §3 | Par: {The same source coordinate characterization} | f:1 | Hyp: source hypotheses | Stat: P | Use: Provides positive cases alongside the nonlattice theorem.

**Statement.** For a nonempty finite family of static machines, put  $c = \min_i h_i$  and  $d = \min_i u_i$ . If  $c + d$  is a positive power of two, the meet within the static subclass is  $S(c, d)$ .

**Proof.** It is an allowed lower bound. Every other lower bound has both coordinates at most  $c, d$ , so lies below it.

#### 168. LF-A5.3. Logical horizon monotonicity does not imply heuristic runtime monotonicity

LF-A5.3 | Src: V48 Int. B §§4–5 | Par: {} | f:0 | Hyp: stated conditions + declared background | Stat: P | Use: Keeps proof geometry separate from implementation effort.

**Statement.** Even though  $hJ(k)$  is nonincreasing, a fixed heuristic may run slower after BANK growth because branching, indexing, duplicate work, or memory pressure can increase.

**Proof.** The shortest available route and the route actually discovered by a policy are different quantities.

#### 169. LF-A6.2. Logical-round dominance does not imply wall-clock dominance without resource replication

LF-A6.2 | Src: V48 Int. B §§6–7 | Par: {Source Theorem 6.3 and Remark 6.4} | f:1 | Hyp: source hypotheses | Stat: P | Use: Prevents a scheduling theorem from being overread as a hardware-speed theorem.

**Statement.** The emulation inequality can fail as a wall-clock statement if  $P$  loses processor time, verifier service, memory, or scheduling priority relative to the stand-alone prover.

**Proof.** Theorem 6.3 counts local opportunities, while Remark 6.4 lists resource-sharing mechanisms that break the wall-clock comparison.

#### 170. LF-IIT22.3. Equal self-theories need not force equal mechanism properties

LF-IIT22.3 | Src: Core II §22 | Par: {LF-IIT22.1} | f:1 | Hyp: source hypotheses | Stat: P | Use: Gives a negative reconstruction result: theorem content alone need not recover full mechanism structure.

**Statement.** There can be two presented mechanisms that agree on the relevant self-theory data while differing on a mechanism property covered by LF-IIT22.1.

**Proof.** If self-theory determined every such mechanism property, the separation asserted by LF-IIT22.1 would be impossible.

#### 171. LF-D18.4. Formal self-awareness level is not forced by computational complexity alone

LF-D18.4 | Src: Core II §18 | Par: {LF-D18.3} | f:1 | Hyp: source hypotheses | Stat: P | Use: Prevents complexity from being used as a proxy for the formal level definition.

**Statement.** No implication from the source complexity criterion alone to positive formal self-awareness can hold in this framework.

**Proof.** LF-D18.3 supplies a counterexample satisfying the complexity side while having level zero.

## 172. LF-21.2. Dependency depth is bounded by inference-node count

LF-21.2 | Src: V49 §21 | Par: {} | f:0 | Hyp: declared definition/background | Stat: P | Use: Provides a consistency check without equating depth and size; branching and sharing can separate them.

**Statement.** A finite certificate DAG with  $q$  primitive inference nodes has maximum primitive inference depth at most  $q$ .

**Proof.** A directed path in a DAG cannot repeat a node. Hence every such path contains at most the total number  $q$  of inference nodes.

## 173. LF-A1.3. Metalogical records cannot be silently used as object-level premises

LF-A1.3 | Src: V48 Int. B §§1–3 | Par: {} | f:0 | Hyp: declared definition/background | Stat: P | Use: Prevents a type error from masquerading as cooperative deduction.

**Statement.** A D- or I-channel certificate stored in the metalogical BANK is not an F -premise unless a separately verified object-level formula is deposited into the object BANK.

**Proof.** The architecture declares the BANK components type-distinct, so membership in one does not imply legal use in the other.

## 174. LF-ML1.5. A legal path weight is zero only when a step has zero policy weight

LF-ML1.5 | Src: V48 Int. A | Par: {} | f:0 | Hyp: declared definition/background | Stat: P | Use: Distinguishes low probability from outright policy exclusion.

**Statement.** For the idealized product  $p_{\theta,T}(\pi) = \prod \text{constituent step has positive weight}$ .  $\prod_j p_{\theta,T}(a_j | S_j)$ , the path has positive weight whenever every

**Proof.** A finite product of positive real numbers is positive. Standard softmax assigns positive weight to every finite-score legal move.

## 175. LF-A1.1. A proof or refutation already implies dependence

LF-A1.1 | Src: V48 Int. B §§1–3 | Par: {} | f:0 | Hyp: declared definition/background | Stat: P | Use: Shows why the D channel is an architectural role, not a fifth exclusive truth value.

**Statement.** If  $B \vdash F T$  or  $B \vdash F T \neg$ , then  $\text{Dep}_{F,B}(T)$  holds.

**Proof.** Dependence is defined as the disjunction of those two derivability predicates.

## 176. LF-A5.2. A shorter explicit-premise horizon need not shorten the fully expanded proof

LF-A5.2 | Src: V48 Int. B §§4–5 | Par: {} | f:0 | Hyp: stated conditions + declared background | Stat: P | Use: Prevents macro compression from being misreported as an unconditional primitive-proof speedup.

**Statement.** A BANK lemma can reduce  $\|T\|_F(B_k)$  while leaving the primitive proof from  $B_0$  just as long once the lemma's own certificate is expanded.

**Proof.** The explicit-premise measure treats deposited lemmas as premises; full expansion charges the derivation used to create them.

## 177. LF-22.1. One useful deposited intermediate suffices in the toy system

LF-22.1 | Src: V49 §22 | Par: {} | f:0 | Hyp: stated conditions + declared background | Stat: P | Use: Exhibits redundant additional BANK material for a fixed target, rather than assuming every extra deposit strictly improves its horizon.

**Statement.** For the exact rule set  $r_1(a, b) = c$ ,  $r_2(c) = d$ ,  $r_3(b) = e$ ,  $r_4(e) = d$  and  $B_0 = \{a, b\}$ , adjoining either  $c$  or  $e$  reduces the inference horizon of  $d$  from two to one. Adjoining both still gives horizon one; adjoining  $d$  itself gives zero.

**Proof.** Initially neither rule concluding  $d$  has its required premise, and each missing premise can be obtained in one step, so exactly two steps are necessary and sufficient. Either  $c$  or  $e$  enables a one-step final rule. With both,  $d$  is still not an initial premise, so zero steps are impossible. With  $d$  as a premise, zero steps suffice.

## 178. LF-ML1.4. Equal scores produce a uniform legal-move policy

LF-ML1.4 | Src: V48 Int. A | Par: {} | f:0 | Hyp: declared definition/background | Stat: P | Use: Provides a neutral reference policy inside the same legal action set.

**Statement.** If  $s_{\theta}(S, a)$  is the same constant for every  $a \in A(S)$ , then  $p_{\theta,T}(a | S) = 1/|A(S)|$  for every legal move.

**Proof.** All exponentials in numerator and denominator are equal, so each gets one equal share of the total.

## 179. LF-ML2.3. A fixed-instance finite-budget win is not an asymptotic speedup theorem

LF-ML2.3 | Src: V48 Int. A | Par: {} | f:0 | Hyp: stated conditions + declared background | Stat: P | Use: Keeps empirical proof-of-concept evidence within its declared scope.

**Statement.** The reported pcom separation alone cannot establish an asymptotic speedup claim for a target family.

**Proof.** An asymptotic theorem requires a scaling family and proved performance bounds as the size parameter grows; the source run supplies one right-censored finite instance.

## 180. LF-ML2.2. A discovery-cost separation is not a proof-length separation

LF-ML2.2 | Src: V48 Int. A | Par: {} | f:0 | Hyp: stated conditions + declared background | Stat: P | Use: Prevents performance evidence from being restated as a proof-complexity theorem.

**Statement.** Different expansion counts for two searches do not imply that their emitted verifier-accepted certificates have different proof lengths.

**Proof.** Expansion discovery cost counts search states; proof length counts the certificate under a separate convention. Neither definition determines the other.

## 181. LF-A1.2. Timeout cannot certify independence

LF-A1.2 | Src: V48 Int. B §§1–3 | Par: {} | f:0 | Hyp: declared definition/background | Stat: P | Use: Protects the verifier boundary against converting failed search into metatheory.

**Statement.** Failure of P and R to find certificates within a finite budget does not establish  $\text{IndF,B}(T)$ .

**Proof.** Independence requires both nonderivability statements, whereas timeout records only absence of discovery within the observed search budget.

## 182. LF-ML1.3. Softmax legal-move probabilities normalize

LF-ML1.3 | Src: V48 Int. A | Par: {} | f:0 | Hyp: declared definition/background | Stat: P | Use: Checks that the displayed learned weighting is a probability distribution on the frozen legal action fiber.

**Statement.** For finite nonempty  $A(S)$  and  $T > 0$ , the idealized softmax weights satisfy  $\sum_{a \in A(S)} p_{\theta,T}(a | S) = 1$ .

**Proof.** The common denominator is exactly the sum of all numerators  $\exp(s_{\theta}(S, a)/T)$  over  $A(S)$ .

Appendix: source-volume provenance

- Volume I: 50 entries - Version 49 integrations, Sections 1-35.
- Volume II: 35 entries - Corrected Core Part I.
- Volume III: 36 entries - Induction and Early Formal Self-Awareness.
- Volume IV: 27 entries - Discovery, Picard, Settlement, IIT, and Computability.
- Volume V: 34 entries - Learned Search and Cooperative AMLD.

All statements and written proofs in this edition are reorganized from the five supplied Low-Fan-In Encyclopedia volumes. The strength ordering is editorial; the stable theorem IDs and source provenance are preserved so readers can recover the original section organization.

Suggested arXiv taxonomy reference: [https://arxiv.org/category\\_taxonomy](https://arxiv.org/category_taxonomy)